

Propose, Don't Judge: An Anytime-Valid Referee for LLM Agents That Mine Investment Factors

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Abstract

Language-model agents now run the whole of quantitative factor research: they propose investment factors, backtest them, select the survivors and retire them. We ask which of those jobs an agent should keep. Our answer is *governed self-evolution*: the agent may propose, and a frozen statistical referee that the agent cannot touch must judge. The referee scores each candidate only on market outcomes revealed after submission, by betting, so its false-discovery guarantee holds at every stopping time for any proposal policy. We cross three proposers (a script, a bandit and a language model) with this referee and with three deliberately leaky ones, in a synthetic world with planted truth, a probe-authoring environment and a ten-year walk-forward on the CSI 500. Who judges sets the number of false admissions: the frozen referee admits $5\text{--}11\times$ fewer sub-threshold factors than the leaky referees under a scripted proposer, and no proposer closes that gap. Who proposes sets the yield: the language model beats the script, matches the bandit, and adds the one capability a bandit lacks, writing its own diagnostic probes. The certificate's price is time: an admitted true factor waits about 500 trading days, and the certified portfolio's Sharpe ratio therefore trails an ungated one. Judging belongs to the procedure; proposing and instrument-making belong to the agent.

1 Introduction

Language-model agents are taking over quantitative factor research. By our count, about twenty-five agentic factor-mining systems appeared in the eighteen months to mid-2026. In every system we examined, the agent performs every job in the pipeline. It generates candidate factors, backtests them, selects the survivors, and decides when to drop them. This paper asks which jobs the agent should keep. We answer by measuring a division of labour. The agent may propose. A statistical procedure that the agent cannot touch must judge.

This division matters because evidence about a factor arrives slowly. An *investment factor* scores every stock on a characteristic and bets that the ranking predicts returns. The bet buys the top fifth, sells the bottom fifth, holds, and repeats. Its daily report card is the *rank-IC*. This is the rank correlation between yesterday's ranking and today's returns across the universe. In a 500-stock universe, a good factor averages two or three hundredths a day. Its standard deviation is several times larger. A factor therefore earns trust only over years. A search over hundreds of candidates finds many that look good by luck.

An agent in a loop creates two errors that today's evaluations were not designed to control. The systems we examined apply a fixed checklist to the agent's output: a *t*-statistic, an information ratio, or, at best, a deflated Sharpe ratio. Those checklists assume a batch of tests fixed in advance. An agent is not a fixed batch if it reads each verdict before choosing its next proposal and never stops. It resubmits until something passes. This

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is *optional stopping*. It passes near-duplicates of a lucky candidate together. This is *multiplicity*. A correction computed in advance requires a proposal process known in advance. A learning agent has no such process. No prior system separates proposing from judging. The one line of prior work that puts an anytime-valid test around an agent certifies the agent’s own modifications, not the hypotheses it proposes (Sengupta 2026; Shawn 2026). A survey of self-improving systems reaches the same gap from the other side: every improvement loop rests on a verification signal, and governance-grade measurement of that loop is underdeveloped (Chen et al. 2026). The open question is therefore **which jobs in a factor lifecycle benefit from an agent’s intelligence, and which must be governed by a statistical procedure the agent cannot touch**. We call the resulting arrangement *governed self-evolution*. The agent’s surface may evolve without limit: its proposals, its diagnoses, and the instruments it writes for itself. A frozen statistical referee governs what enters the portfolio. The referee scores a candidate only on market outcomes revealed after submission. We split the pipeline into a *proposer* and a *referee*, then freeze the referee. Each candidate factor is scored on the rank-IC from the days after submission. Nothing the proposer saw before submission enters the score. The score is a bet. Each day, the referee stakes part of the candidate’s capital on that day’s rank-IC being positive. A factor with no edge rarely produces capital that grows large. A process with this property is called an *e-process*.

Two further rules act on that capital. The admission rule asks whether a candidate has any edge. A candidate is *admitted* when its capital clears a bar accounting for every candidate ever submitted. That rule is *online e-BH*, the e-value form of the Benjamini–Hochberg procedure. The retirement rule asks whether a deployed factor’s edge still covers its own trading costs. The registered viability threshold δ is the edge that just pays. A factor is *retired* when a second bet against it shows that its edge has fallen below δ . That rule is an *e-detector*, the e-value form of a changepoint chart. Section 4 builds all three procedures, and Appendix A collects their definitions.

The guarantee holds regardless of who proposes. Every bet is sized before the day’s outcome is seen. The market reveals outcomes regardless of the proposer’s actions. The false-discovery rate over admissions is therefore controlled at every stopping time. The retirement detector’s run length to a false alarm is bounded below. Both statements hold for any proposer, including one that rewrites its own agent surface (§4.5). They rely on one hypothesis: the proposer has no information about the post-submission stream. A live run satisfies this hypothesis. A historical replay by a model trained on the period may not.

The proposer can therefore be anything, and we test a ladder of three. A *round-robin script* cycles through the factor families in a fixed order. A *discounted-UCB bandit* is the standard multi-armed-bandit allocator. It sends more submissions to families with recent successes and slowly forgets. A *language-model controller* also allocates submissions. It diagnoses retirements and writes its own *perception probes*. These small diagnostic programs measure a suspected fault and never enter a statistical test.

The experiment separately measures the effects of who judges and who proposes. Besides the valid referee, we run three deliberately leaky referees that each mimic a common practice. Crossing the three proposers with the four referees measures each effect while holding the other fixed. Three environments share this design. A synthetic world with planted truth isolates the mechanism. A probe-authoring environment with held-out fault types measures the one agent capability we claim. A ten-year point-in-time walk-forward on the CSI 500 tests which results survive real data. It also follows the certified factors into a portfolio after costs. Every comparison uses the same random draws, and model families are never pooled.

What we find. A *campaign* is one walk-forward run of the pipeline, lasting seven to ten years depending on the start year. Real-data counts are per campaign. A *false admission* is an admitted factor whose realised edge falls short of the viability threshold.

First, who judges sets the number of false admissions, and no proposer substitutes for it. In the synthetic world, the frozen referee admits no false factor in any seed. The three leaky referees admit 0.26–0.85 per submission. The figures are the same for script, bandit, and language model (Table 8). On real data, the frozen referee admits 11.7 false factors per campaign. The leaky referees admit 86–196 under the script and 37–78 under the language model. The frozen referee’s advantage is $5\text{--}11\times$ under the script and $2.6\text{--}4.9\times$ under the language model, based only on days after admission. A patient desk that waits 500 days and then corrects for multiplicity comes close where families are clearly good or bad. It is four times less accurate where many edges are small (§8.5).

Second, who proposes sets the yield, and the language model’s value is real and bounded. The language model finds more true factors than the script in six of six family $\times$ library settings, in every start year. It is level with the bandit in most settings. It falls behind in one family on the nine-family library. Its one capability a bandit lacks is instrument-making. Authored probes significantly lower intervention regret in three of six evaluable model families. In the other three, they are indistinguishable from a fixed diagnosis menu, so authoring never hurts (Table 9).

Third, the certificate has a price, paid in waiting and in what the bar selects. The certified portfolio does not lose money at the registered cost. Its alpha cannot be told from zero. Its Sharpe ratio net of costs is below the ungated portfolio’s. Two mechanisms account for the gap. An admitted true factor waits about 500 trading days. The ungated pipeline is invested during that wait. The strongest daily statistic clears the bar fastest. In this library, that statistic is short-horizon reversal, the family that does not pay after costs. A fixed-horizon test with the same patience therefore builds a better portfolio (§8.7).

Contributions. We propose and measure governed self-evolution, a framework pairing an evolving agent surface with a frozen statistical referee.

- (C1) *A referee that can be frozen.* It scores each candidate only on data revealed after submission. The false-discovery rate over admitted factors therefore holds at every stopping time, whatever the proposer does. We state the conditions under which this holds inside an agentic loop.
- (C2) *An experiment that separates the two jobs.* Three proposers each run against the valid referee and three leaky ones. Common random draws let the two effects be read separately.
- (C3) *A measurement of what the language model adds.* It beats the script on yield. It does not reliably beat the bandit on allocation. It writes its own diagnostic probes for faults it has never seen.
- (C4) *A ten-year real-data study on the CSI 500* that follows the certified factors into a portfolio with costs. It accounts for the gap between the certified and ungated portfolios.

We do not offer an investment system. The portfolio layer is a measuring instrument. The paper has three stages: the pipeline (§4–§6), the experiment (§7) and what it found (§8–§9).

2 Related work

Agentic factor mining treats the agent as the whole pipeline. Research spans reinforcement-learning generators over an operator grammar (Yu et al. 2023), agentic loops with critique and refinement (Li et al. 2025; Tang et al. 2025), and structured-semantics search (Yi et al. 2026). In each approach, the same agent generates, evaluates, and selects. Evaluation uses held-out IC, ICIR, or Sharpe after a fixed mining budget. At most, a static deflation is applied to a batch (Bailey and López de Prado 2014). None controls the sequential error created by the loop. The closest evidence is a boundary finding. AlphaSchema reports that model identity has little effect within a structured search (Yi et al. 2026). Our proposer ladder reproduces that finding.

Anytime-valid inference supplies every piece the referee needs. We apply it without extending it. Test supermartingales and Ville’s inequality (Ville 1939) permit a test that can be read every day. We use betting e-processes for bounded means (Waudby-Smith and Ramdas 2024). The e-BH procedure (Wang and Ramdas 2022) and its online form (Fischer et al. 2024; Wang et al. 2025; Heide 2026) control the false-discovery rate over a growing set at data-dependent stopping times. E-values merge under arbitrary dependence (Vovk and Wang 2021). Compound and weighted e-values (Xu et al. 2026; Sun and Wang 2026; Clerico 2026) fix the form of the bet. Our contribution states when this theory survives an agentic loop. It also measures the cost. Certified self-evolution is the nearest architecture, but it certifies a different object. SEA (Sengupta 2026) and PACE (Shawn 2026) gate a self-modifying agent with an anytime-valid test. Both certify *the agent’s own modifications*. A factor pipeline is harder. The agent proposes external hypotheses whose truth a market decides. The admitted set grows without bound. The guarantee must control the false-discovery rate over that set. The division of labour, the pipeline implementing it, and the experiment measuring it are ours.

Retirement and controller comparison each draw on established results. Factor edges decay after discovery (McLean and Pontiff 2016). Many published factors are false from the outset (Harvey et al. 2016). A deployed factor must therefore be monitored. E-detectors (Shin et al. 2024) provide our retirement rule. Error-over-patience (Dandapanthula and Ramdas 2025) is the library-level metric. Static replay of logged agent trajectories is invalid once policies diverge (Gonuguntla 2026). Controllers are therefore compared through branched live rollouts with common random numbers. Portfolio results are reported after costs. They also use the deflated Sharpe ratio (Bailey and López de Prado 2014). Table 1 summarises what each line certifies.

Table 1: What each line of work certifies and when its guarantee holds. “Adaptive proposals” are candidates a learning agent chooses using earlier candidates’ outcomes. “Growing set” means controlling the false-discovery rate over every factor ever admitted. The agentic miners are (Yu et al. 2023; Li et al. 2025; Tang et al. 2025; Yi et al. 2026). Certified self-modification is (Sengupta 2026; Shawn 2026). Notes: ^a denotes a fixed batch. ^b denotes a bounded modification space. ^c denotes a per-modification budget. ^d denotes online e-BH. For ^e, the e-detector guarantees the average run length (§4.4). ^f applies to a proposer with no information about post-submission outcomes (§4.5).

Approach	Certifies	Anytime Valid	Adaptive Proposals	Growing Set	Retirement
Backtest Checklist	A batch of backtests	No	No	No	No
Batch Deflation	A batch of trials	No	Partial ^a	No	No
Agentic Miners	As the two rows above	No	No	No	Heuristic
Certified Self-Modification	The agent’s own modifications	Yes	Partial ^b	No ^c	No
This Paper	External hypotheses, scored after submission	Yes	Yes ^f	Yes ^d	Yes ^e

3 Problem: an adaptive, feedback-coupled factor stream

This section defines the objects used throughout the paper. They are the candidate stream, market outcomes, candidate states, guarantees, outcome measures, and threats.

We study a stream of candidate factors whose next member depends on everything observed so far. A library $\mathbb{L}$ is a finite set of factor families $f \in \mathcal{F}$ and their parameter grids. Each family represents an economic hypothesis, such as momentum or value. A proposer, called the controller, emits candidate factors $\varphi_1, \varphi_2, \dots$

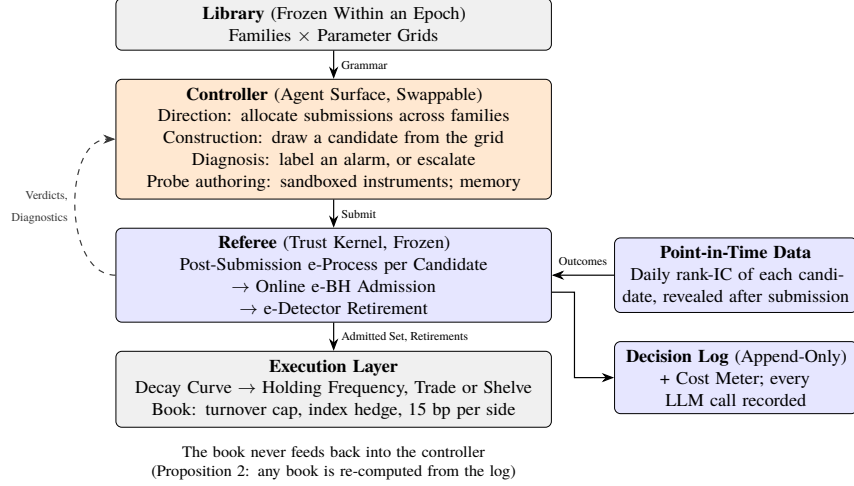


Figure 1: The division of labour. The controller (orange) proposes, diagnoses and writes its own instruments. The referee, the data layer and the decision log (blue) are frozen components the controller cannot mount, patch or intercept. The execution layer (grey) turns admitted factors into positions and is never seen by the referee or the controller.

Candidate i is a pair (f_i, θ_i) submitted on trading day s_i . Candidates within a family are near-duplicates. An adaptive controller resubmits variants of candidates that passed. The stream is therefore neither independent nor fixed in advance.

The market reveals outcomes one day at a time. Let $\mathcal{U}_t$ be the point-in-time universe on day t . It contains stocks whose index membership was known at $t - 1$ and that are tradable and not limit-locked. For day $t > s_i$, candidate i 's outcome is $X_t^{(i)}$. It is the cross-sectional Spearman rank correlation between its signal at the close of $t - 1$ and the returns on t , over $\mathcal{U}_t$. Outcomes are exogenous. No action on day t changes $X_t^{(i)}$.

Anything used on a day must have been known the day before. This rule has a formal name. Write $\mathcal{G}_{t-1}$ for the *global filtration*. It contains everything known at the close of day $t - 1$, including all market data, candidates' outcomes, controller outputs, and referee states. Every quantity used on day t must be measurable with respect to $\mathcal{G}_{t-1}$. We call this rule the alignment invariant (§4.5).

A candidate passes through three states, shown in Figure 1. Only the referee moves it between them. A candidate is *open* until it is admitted or expires. An admitted candidate is *deployed*. The detector makes a deployed factor *retired*. The portfolio, called the *book*, holds every deployed factor.

The referee provides two guarantees. Both must hold at any horizon and any stopping time. The false-discovery rate over admissions is at most α . The expected run length to a false retirement is at least A . Appendix A defines a false retirement and its clock.

Two outcome measures record what a campaign produced. To compute *Realised yield per submission*, count factors still deployed at campaign end whose post-submission mean rank-IC is at least δ . Divide this count by the number of candidates submitted. *Realised false admissions* counts factors ever admitted whose post-submission mean rank-IC falls short of δ . Real data have no planted truth. We therefore label a candidate true when its mean rank-IC from submission to campaign end is at least δ . The candidate and its submission day fix the label. The label is blind to the referee's decision.

Two properties of that label determine how to read the real-data counts. First, the label's window starts at submission. It therefore includes the days on which the referee decided. This overlap favours a referee that admits late. To remove it, §8.4 re-judges every admitted factor using only the days after its admission. Second,

a realised false admission is a count against the viability threshold δ . The guarantee bounds a false-discovery *proportion* against a *zero* edge (§4.2). An admitted factor with a small positive edge is therefore a realised false admission, not an error of the guarantee.

The threat model contains four entries. The referee addresses three by construction. Optional stopping and resubmitting near-copies are the two errors of §1. *Verifier-in-the-loop adaptation* means that the controller learns what the referee rewards. The fourth threat is *knowledge-cutoff leakage*. A live run removes it because no proposer can know the post-submission stream. A historical replay may not remove it if the model’s training covers the campaign years. The guarantee’s hypothesis excludes that case (§4.5). The language-model arms’ real-data results are therefore read as replay outcomes. Appendix A maps each threat to its answer.

4 The referee

The referee answers three questions about every candidate factor. Does this candidate have any edge at all? Which candidates can be admitted while keeping false admissions low among all candidates ever submitted? Once deployed, has a factor’s edge fallen below the level that pays for its trading?

A classical test cannot answer them here. It fixes its sample size in advance and is read once. The referee is read every day over a candidate set that grows as long as the system runs.

One idea answers all three questions: treat evidence as capital in a betting game. The game is fair or unfavourable to the candidate whenever the candidate is in fact null. Capital in such a game cannot be expected to grow. Therefore, the chance that it ever reaches a high level is small, whenever one looks. The first question becomes a bet that one candidate’s edge is positive. The second becomes a rule for how high each candidate’s capital must climb when many candidates are betting. The third becomes a bet against a deployed factor. Its capital grows only when the edge is gone.

The section follows that order. Section 4.1 states the one property of betting capital underlying all three procedures. Sections 4.2, 4.3 and 4.4 build those procedures. Figure 2 shows one candidate’s life under them. Section 4.5 gives the condition under which an agent in the loop leaves all three intact.

4.1 Evidence as capital

A test supermartingale measures evidence and stays honest when read every day. Write $(\mathcal{G}_t)$ for the global filtration of §3, containing everything known at the close of day t . A nonnegative process W with $W_0 = 1$ is a *test supermartingale* for a null hypothesis H_0 if $\mathbb{E}_{H_0}[W_t \mid \mathcal{G}_{t-1}] \leq W_{t-1}$. Under the null, tomorrow’s expected capital is at most today’s.

Two consequences support everything that follows. The first is *optional stopping*: $\mathbb{E}_{H_0}[W_\tau] \leq 1$ for any stopping time τ . Such a stopping time is any rule that uses only the past to decide when to stop. Capital read at a data-dependent moment remains a fair quantity, called an *e-value*. The second is *Ville’s inequality* (Ville 1939): $\mathbb{P}_{H_0}(\exists t : W_t \geq 1/\alpha) \leq \alpha$. A null candidate’s capital reaches $1/\alpha$ with probability at most α .

4.2 One candidate: a bet on the post-submission stream

The first procedure tests whether one candidate has an edge. It plays the game of §4.1 on that candidate’s post-submission stream. Candidate i ’s outcomes are its daily rank-IC $X_t \in [-1, 1]$ on the days after submission. The null says it has no edge: $H_0 : \mathbb{E}[X_t \mid \mathcal{G}_{t-1}] \leq 0$. Given everything known the day before, its expected rank-IC is at most zero. Each day, the referee stakes a fraction λ_t of the candidate’s capital on a positive

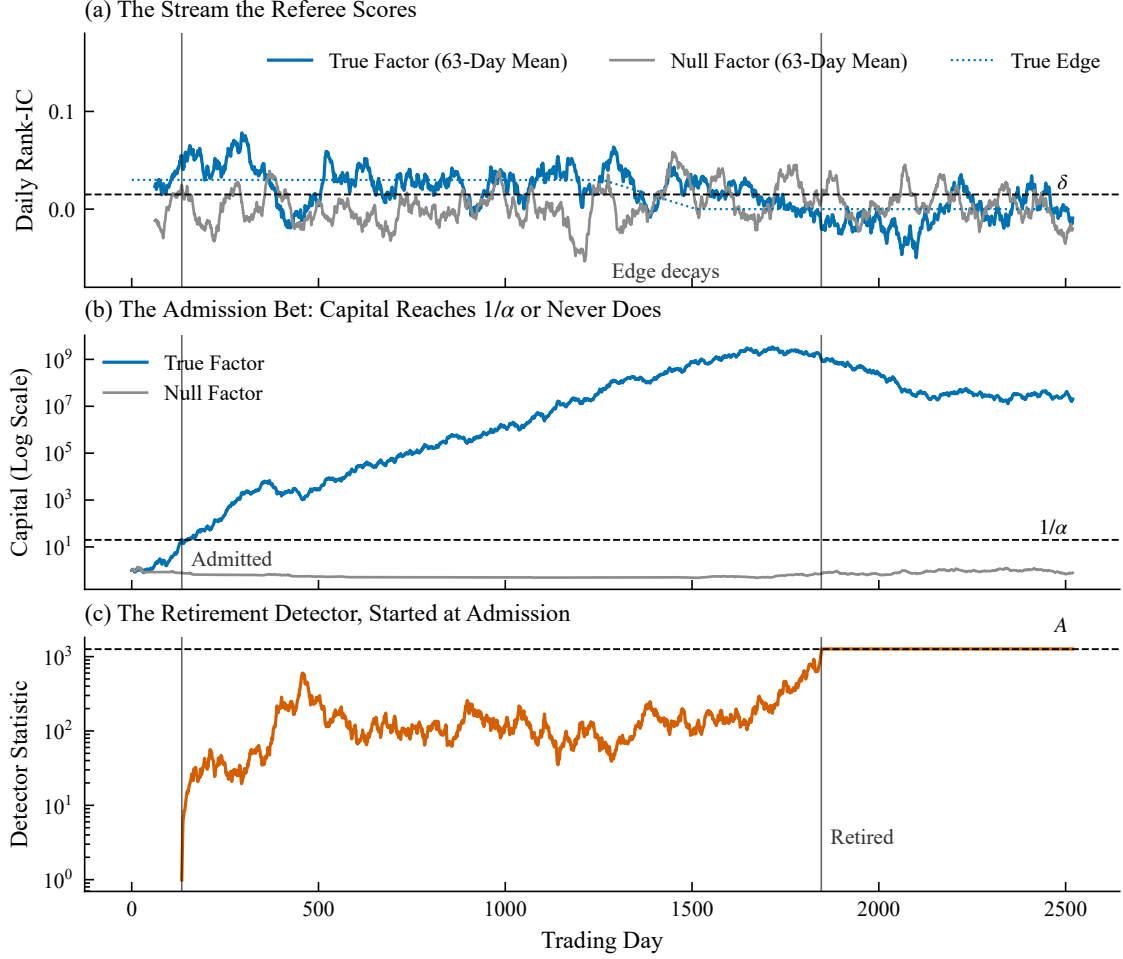


Figure 2: One candidate’s life under the referee, on a simulated stream (an illustration of the mechanics, not a reported result). Top: the 63-day mean of the daily rank-IC of a factor whose edge is 0.03 and then decays to zero, and of a null factor, with the viability threshold δ . Middle: the capital of the admission bet on a log scale. The true factor’s capital reaches the single-candidate bar $1/\alpha$ drawn here (the bar as run, over N_v slots, is higher, §4.3) and the factor is admitted; the null factor’s capital never does. Bottom: the retirement detector, started at admission, stays flat while the edge is intact and climbs to A once the edge has decayed.

outcome,

$$W_t = \prod_{s \leq t} (1 + \lambda_s X_s), \quad \lambda_s \in [0, \lambda_{\max, s}] \text{ } \mathcal{G}_{s-1}\text{-measurable, } \lambda_{\max, s} < 1.$$

The referee fixes λ_t before seeing X_t . The null makes the expected value of X_t at most zero. These conditions make W a test supermartingale. The cap on λ keeps every product factor positive. The capital therefore can never be lost outright.

The bet is fair to a null candidate, however cleverly it was chosen, under one proviso. The proposer’s choice must use no information about the post-submission stream. This condition holds in a live run, whatever the proposer learned from the past. It may fail in a historical replay. A model may hold such information if its training window covers the stream. For the language-model arms, the guarantee therefore applies to live deployment. The replay results are outcomes (§9).

Every feasible bettor is slow at realistic noise levels because noise sets the speed. An alternative with conditional mean $\mu > 0$ and variance σ^2 has expected daily log-capital growth of about $\lambda\mu - \lambda^2\sigma^2/2$. Growth is largest at the Kelly stake $\lambda^* = \mu/\sigma^2$. Capital then reaches $1/\alpha$ after roughly $\ln(1/\alpha) \cdot 2\sigma^2/\mu^2$ days. Daily rank-IC on a 500-stock universe has $\sigma \approx 0.12$. For relevant edges ($\mu = 0.02$ – 0.03), the Kelly stake exceeds the no-bankruptcy cap. We use the plug-in aGRAPA stake (Waudby-Smith and Ramdas 2024), capped at a fraction $\phi = 0.8$ of that bound:

$$\lambda_t = \text{clip}\left(\frac{\hat{m}_{t-1}}{\hat{v}_{t-1} + \hat{m}_{t-1}^2}, 0, \lambda_{\max,t}\right),$$

Here, $\hat{m}_{t-1}$ and $\hat{v}_{t-1}$ are the running mean and variance of X_s over $s < t$. The cap fraction, not the bettor, sets the wait (Appendix B).

Autocorrelation changes which null makes the bet fair, so the referee whitens the stream. A factor with a *marginal* edge of exactly zero has a positive conditional mean under AR(1) noise after a positive shock. A bet that conditions on the past exploits those days. The raw-stream referee admits 6.0% of such factors over ten years at $\rho = 0.2$ and 15.4% at $\rho = 0.4$, against a nominal 5%. The referee therefore bets on the *predictably whitened* stream

$$\tilde{X}_t = X_t - \hat{\rho}_t X_{t-1}, \quad \hat{\rho}_t = \text{shrunk, capped estimate from } X_{<t}, \quad \lambda_{\max,t} = \frac{\phi}{1 + \hat{\rho}_t},$$

against $\mathbb{E}[\tilde{X}_t \mid \mathcal{G}_{t-1}] \leq 0$. Every quantity is $\mathcal{G}_{t-1}$ -measurable. The supermartingale argument therefore holds for this null. The theorem gives no finite-sample bound for the marginal null. Thus, control of the marginal edge is measured, not proved. Whitening restores false-admission rates of 1.8, 2.0 and 2.4% at $\rho = 0, 0.2, 0.4$ (Table 7, Appendix B).

Admission certifies an edge. Whether that edge pays for its own trading is a separate question. The registered threshold δ addresses that question. A factor with edge μ earns about $\kappa\mu$ per day gross. Here, κ converts one unit of daily IC into return. It pays $2c \cdot \text{TO}$ per day for per-side cost c and one-way turnover TO. The break-even edge is

$$\delta = \frac{2c \cdot \text{TO}}{\kappa}.$$

κ is the *average* conversion from daily IC into gross return. We use 0.018 (§5.2). The break-even differs across families by an order of magnitude because turnover does. At $c = 15$ bp, it is 0.010 for value, 0.012 for liquidity, 0.017 for volatility, 0.031 for momentum and 0.074 for short-horizon reversal.

One δ across these families is therefore a *signal-quality floor*, not a per-family economic threshold. It was registered as a single value, $\delta = 0.015$, before the runs. It is the null of the retirement detector. It defines the realised label of §3. It is the execution objective at the shortest hold (§5). The execution layer decides whether a certified factor covers its own costs.

4.3 Many candidates: online e-BH over a frozen universe

The second procedure decides which candidates to admit when many are available. The rule is Benjamini–Hochberg with capital instead of p -values. Its objective is the false-discovery rate over everything ever admitted. The universe is frozen at N_v slots per epoch. Each candidate occupies its own slot with weight

$\gamma_c = w_v/N_v$, $\sum_v w_v \leq 1$. The admitted set on day t is

$$R_t = \left\{ c : E_c(t) \geq \frac{1}{k_t^* \alpha \gamma_c} \right\}, \quad k_t^* = \max \left\{ k : \#\{c : E_c(t) \geq \frac{1}{k \alpha \gamma_c}\} \geq k \right\},$$

where $E_c(t)$ is the candidate's capital, frozen at its admission-day value (Fischer et al. 2024). More candidates clearing a bar lowers the bar each faces. Appendix A works a two-candidate example.

The guarantee requires no assumption about dependence between candidates. If c is admitted, $\mathbf{1}\{c \in R_t\}/|R_t| \leq \alpha \gamma_c E_c$, which is $\alpha E_c/N_v$ at $w_v = 1$. Taking expectations under each null and summing over the null candidates gives $\text{FDR} \leq \alpha$ (Wang and Ramdas 2022). The guarantee holds at every stopping time because stopped capital is an e-value. Frozen values never fall, so admissions are never revoked (Wang et al. 2025; Heide 2026).

In the runs, the bar is far above the single-candidate $1/\alpha$. We set $\alpha = 0.05$ and $w_v = 1$, using one epoch with $N_v = 2000$ slots. A campaign uses about 490 slots. An expired or retired candidate keeps its slot, so every admission draws on the same budget. The first admission requires capital of N_v/α . The k -th requires $N_v/(k\alpha)$.

The bar's height sets the wait. The noise determines that wait. Combining the bar with the growth rate of §4.2 gives the expected wait for the k -th admission,

$$T \approx \frac{\ln(N_v/(k\alpha))}{\lambda\mu - \lambda^2\sigma^2/2}, \quad T \approx \ln(N_v/(k\alpha)) \cdot \frac{2\sigma^2}{\mu^2} \text{ at the Kelly stake.}$$

At the run parameters, the first admission of a factor with edge 0.03 takes about 546 days. At the viability threshold, edge 0.015, it takes 1434 days, versus 1356 for an uncapped Kelly bettor. The cap costs almost nothing near the threshold (Appendix A works the three cases). Thus, no better bettor can shorten the wait near the threshold. A factor whose edge sits at δ needs more than five years of evidence. A campaign lasts seven to ten.

Two consequences of the rule shape the rest of the paper. A resubmitted near-copy opens a new candidate with its own slot. Repetition therefore cannot raise the false-discovery rate above α . It only spends the proposer's slots. A weak submission does not dilute the other candidates. The proposer's scarce resources are therefore slots, compute and calendar time. A bandit allocates these scalar quantities well (§6). The engine can also merge near-duplicates into a cluster (Vovk and Wang 2021). The reported experiments do not merge.

4.4 Retirement: the e-detector

The third procedure decides when a deployed factor's edge has gone. A desk needs a rule that rarely retires a healthy factor but reacts within months when the edge is really gone. Real decay is gradual (McLean and Pontiff 2016), so its start has no ground truth.

The detector plays the betting game of §4.1 *against* the factor. It uses the viability threshold as the null. The retirement null says the factor remains viable on the whitened stream: $H_0^{\text{ret}} : \mathbb{E}[\tilde{X}_t \mid \mathcal{G}_{t-1}] \geq \delta_t$ with $\delta_t = \delta(1 - \hat{\rho}_t)$. The test bets the other way, on $Y_t = \delta_t - \tilde{X}_t$. Capital grows only while the edge is demonstrably below the floor. A new e-process $W^{(j)}$ starts every day j after admission because decay can begin on any day. The statistic is their sum:

$$M_t = \sum_{j \leq t} W_t^{(j)}, \quad \tau_A = \inf\{t : M_t \geq A\}.$$

The restarts make the detector responsive. A single e-process started at admission would dilute new evidence with the factor’s healthy period.

The guarantee concerns the average run length. Each $W^{(j)}$ is a test supermartingale under the retirement null, so $M_t - t$ is a supermartingale. Optional stopping then gives $\mathbb{E}_{H_0}[\tau_A] \geq A$ for any A , nonasymptotically and without i.i.d. assumptions (Shin et al. 2024). The same inequality has the finite-horizon form $\mathbb{P}(\tau_A \leq t) \leq t/A$. It is vacuous over a ten-year campaign at $A = 1260$. The false-retirement rates below are therefore simulation results, not consequences of the bound.

The stake targets the decay one wants to catch and is not fitted to the data. A stake fitted to a healthy factor’s drift is near zero. Every capital then stays near one, $M_t \approx t$ climbs linearly, and every healthy factor alarms within a bounded horizon. It is the Kelly bet against the design alternative “full decay to zero edge”,

$$\lambda_t = \text{clip}\left(\frac{\delta_t}{\hat{\sigma}_{t-1}^2 + \delta_t^2}, 0, \lambda_{\max, t}\right),$$

with $\hat{\sigma}_{t-1}^2$ the running variance. Each start then has negative expected log growth under the alive null. As a result, M_t plateaus instead of climbing.

The threshold $A^* = 1260$, five years of trading days, was registered before the runs. Closed-form economic designs for memory-type charts are known to fail unchecked (Lorenzen and Vance 1986; Ahmadi-Javid and Ebadi 2021). The harness therefore checks the delay and false-retirement rate by simulation. On synthetic decay, the e-detector’s median delay is 204–322 days by decay shape. This delay matches a Monte-Carlo-calibrated CUSUM. The e-detector needs no calibration step. Its false-retirement rate on ten-year alive streams is 0.5–1.5% at $\rho \leq 0.2$. At $\rho = 0.4$, the rate is 8.5%.

Each factor keeps its own detector and threshold when K factors are monitored at once. *Error-over-patience* (Dandapanthula and Ramdas 2025), the library-level analogue of the false-alarm rate, is bounded by K/A^* under this per-stream rule. Decay shared by a whole family is read from *family detectors*. Each is the mean of the members’ detector statistics M_t . They inform diagnosis and never retire a factor by themselves.

4.5 What an agent in the loop cannot break

The three guarantees survive an adaptive proposer if the proposer cannot reach three things. It cannot reach bet sizing, which happens before the day’s outcome is seen. It cannot reach the scoring data, which all arrives after the candidate was proposed. It cannot change the set of possible hypotheses within an epoch. Under these conditions, the proposer’s behaviour does not enter the argument.

Proposition 1 (validity for any proposal policy). The controller may be any policy (scripted, bandit, LLM-driven or self-modifying) that satisfies four conditions:

- (i) the referee’s betting fractions and cluster weights stay $\mathcal{G}_{t-1}$ -measurable;
- (ii) each candidate is bet on post-submission data only;
- (iii) the library stays frozen within an epoch;
- (iv) each candidate i is chosen with no information about its post-submission outcomes beyond $\mathcal{G}_{s_i-1}$.

Suppose also that each candidate’s null is the one its bettor scores. Every merge must also use weights fixed in advance. The FDR guarantee of admission then holds at every stopping time. The run-length guarantee of retirement holds for every deployed factor.

That hypothesis uses the whitened conditional null of §4.2. In words, a null candidate’s expected whitened rank-IC is at most zero, given everything known the day before.

Proof sketch. Conditions (i) and (ii) make each candidate’s capital a test supermartingale with respect to $\mathcal{G}$. Condition (iv) preserves this property for a candidate selected by the proposer. The selection carries no information about the increments used to score the bet. Weights fixed in advance make a merged capital an e-value. Condition (iii) keeps the hypothesis universe fixed and finite within each epoch. Thus, stopped e-BH (Wang et al. 2025) and dynamic e-closure (Heide 2026) apply. The detector’s guarantee is a per-stream inequality independent of the policy. The proposition follows from the cited results and is not a new theorem. Its value is to rule out one possibility: that an endogenous proposal process breaks validity.

The proposition fails under one named condition, *foretelling*. The engine guards against it. Foretelling means sizing a bet with knowledge of where another candidate’s stream is about to move. Each bet can then be fair on its own information. Yet the merged capital can have expectation above one. The counterexample of (Wang et al. 2025) has $\mathbb{E}[M] = 1.25$. The engine enforces the alignment invariant of §3. A unit test reproduces the counterexample and confirms that the invariant removes it. The invariant cannot verify the conditional-mean assumption itself. The synthetic world tests that assumption where it holds by construction.

The proposition also defines the boundary between the agent and the referee. The trust kernel comprises the referee, the library, the point-in-time data layer, the append-only decision log and the cost meter. These frozen components cannot be mounted, patched or intercepted by the controller. Everything else (probes, memory, allocation and diagnosis policies) is a swappable agent surface. It may evolve freely.

5 The execution layer

The referee certifies a statistic. A book earns money from positions. The execution layer sits between them. Its one parameter sets how long to hold a certified ranking. Its one decision is whether to hold the ranking at all.

5.1 Persistence, and the edge of an h -day hold

How long to hold a ranking depends on how fast its predictive power fades. The referee does not measure this decay. A ranking S_{t-1} known at the close of day $t - 1$ has a *decay curve*

$$IC(k) = \mathbb{E}[\text{rc}(S_{t-1}, r_{t+k-1})], \quad k = 1, 2, \dots$$

where rc is the cross-sectional rank correlation. The referee scores $IC(1)$. $IC(k)$ for $k > 1$ is the same ranking’s value on the k -th day of the hold. A flat curve has the same daily IC at every k . Thus, a daily certificate does not penalise slowness. A steep curve may clear δ yet fail to cover the trading imposed by its decay.

The objective is the net return of a hold. A *sleeve* is a single factor’s own long–short portfolio. It is long the top fifth and short the bottom fifth. A sleeve re-formed every h days earns the curve’s average over the hold. It also pays for re-formation:

$$\text{net}(h) = \kappa \overline{IC}_{1..h} - 2c\tau(h), \quad \overline{IC}_{1..h} = \frac{1}{h} \sum_{k=1}^h IC(k),$$

where $\tau(h)$ is the sleeve’s one-way turnover per day at that frequency. Setting $\text{net}(1) = 0$ recovers $\delta = 2c \text{TO} / \kappa$. Thus, *the viability threshold is the execution objective at the shortest hold*.

5.2 Estimating κ : why the conversion is a market constant

The objective needs the conversion rate κ . We fix it at one market-level value, $\bar{\kappa} = 0.018$. The relevant rate is the *average*, $\mathbb{E}[\text{gross}]/\mathbb{E}[IC]$. It is 0.016–0.019 for the four families that carry the book. The *slope* of daily gross return on daily IC is about three times larger. It measures the rate on high-dispersion days. The average cannot be estimated per sleeve. A per-window ratio divides by a 250-day mean IC whose standard error is the size of a real edge. A first pass that estimated it per sleeve inverted the shelving decisions (Appendix B).

5.3 The decision, and its rate limit

The decision rule is a deliberately slow grid search. The sleeve’s holding frequency is chosen from five desk frequencies. Each is scored on the trailing W days using only inputs known at the decision date. The search runs on first deployment and then at most once every $B = 250$ trading days. An accepted change moves at most one grid step. The sweep over B finds no consistent winner. The pre-stated rule *never worst, least churn* therefore selects 250 (Appendix B).

A factor that cannot pay at any frequency is not traded. If $\max_h \widehat{\text{net}}(h) \leq 0$, the factor stays certified but is *shelved*. This second threshold is deliberately kept apart from δ . The certificate needs one statistic on one daily stream. Tradability is family-specific because $\tau(h)$ differs across families by an order of magnitude.

5.4 Execution neutrality

Proposition 2 (execution neutrality). Let the execution layer be any policy mapping the admitted set and data up to $t - 1$ to positions at t . Then, (i) no guarantee of §4 depends on it. (ii) Any book can be recomputed from the recorded decision log without re-running the agent.

Proof.

- (i) The e-processes depend on candidate streams and predictable stakes. The admission and retirement rules depend on the e-processes. The execution layer is downstream of all of them. It enters no bet, weight or threshold.
- (ii) By construction, the controller’s state Z_t excludes the book (§6). Therefore, the agent’s action sequence depends only on the referee-side history. The book is a deterministic function of that history.

Part (ii) makes the experiment affordable and auditable. The 540 cells of §8 were recorded once, then re-booked under a new execution layer at no cost. All 220 LLM cells reproduced their referee-level metrics exactly.

6 The controller ladder and the perception probe

The three controllers differ in their capabilities, not their inputs. All use the same library, budget, and numeric family statistics. A controller is a policy $\pi(a_t \mid Z_t)$ over actions $\{\text{mine}(f, \theta), \text{respond}(\text{alarm}), \text{probe}, \text{idle}\}$. Its state Z_t includes family statistics, the referee’s diagnostics, open candidates, and paused families. For the LLM, it also includes unstructured context and memory. Each rung adds a capability to the rung below (Table 25, Appendix D).

- **C0 round-robin** ignores Z_t . It cycles through families in a fixed order and replaces each retired family with a new draw.

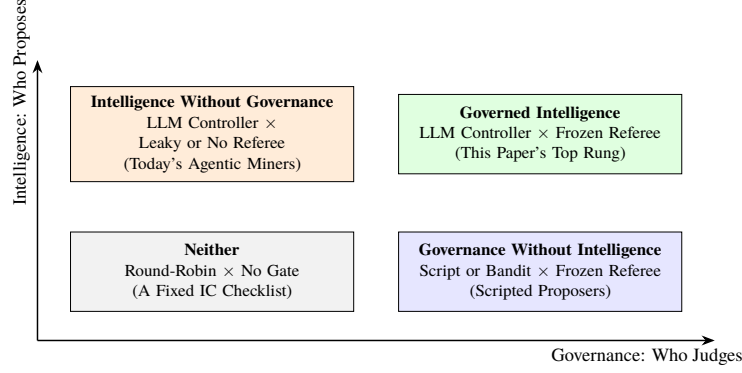


Figure 3: The factorial as a quadrant. Every cell is measured with common random numbers, so the effect of governance (columns) and of intelligence (rows) can be read off separately. The interaction, what the agent adds only when the referee leaks, is a measured quantity.

- **C1 discounted UCB** is a non-stationary multi-armed bandit over families. Each family’s reward is its discounted admission count minus retirements per submission.
- **C2 LLM controller** plays three roles under a fixed scaffold. *Direction*: allocate the next period’s submissions across families using Z_t and each family’s economic profile. *Diagnosis* after a retirement: label the cause or escalate. *Probe authoring*: write, sandbox-test, and mount a perception probe. The probe never enters admission or retirement inference. It therefore carries no multiplicity cost.

The bandit is a strong allocation baseline. This indicates where to look for the LLM’s contribution. Admissions do not dilute one another (§4.3). The direction role therefore allocates scalar resources, which a bandit handles well. The LLM has only two inputs unavailable to the bandit: retirement reasons and family profiles. Any difference must therefore come from diagnosis and instrument-making.

7 Experimental design

7.1 The factorial

The design crosses who judges with who proposes, and every cell is run. Four referee arms span governance. Three controllers span intelligence (Figure 3). Three words recur. An *arm* is one factor setting (a referee, a controller). A *cell* is one run of one arm combination with one model, start year and seed. A *campaign* is a cell’s walk-forward. It lasts seven to ten years, depending on the start year.

The governance axis compares one valid referee with three controlled violations. The *frozen referee* follows the procedure in §4. Each *leaky* arm keeps the rest of the pipeline intact. It breaks one assumption of §4.5 in the way a common practice does:

- **Peeking**: a one-sided t -test uses the post-submission mean. It is checked every day after 60 observations and admits at the first $p \leq \alpha$ (the practitioner’s rolling $t > 1.65$ rule). It breaks optional stopping.
- **Adaptive threshold**: the e-process remains, but the wealth threshold relaxes with candidate age, $\alpha^{-1}e^{-\text{age}/\tau}$. It breaks Ville’s fixed threshold.
- **No gate**: every candidate is admitted after a fixed incubation of 126 trading days, as today’s miners do. It tests nothing.

The leaky arms support two contrasts. Comparing a leaky arm with the frozen arm under one controller

measures what the leak lets through. Comparing two controllers within one leaky arm measures how much apparent adaptive-controller skill comes from the leak.

The intelligence axis has the three controllers of §6. On real data, the LLM controller uses three model families. That quadrant is a replay. A language model trained on the campaign years falls outside condition (iv) of Proposition 1. It therefore reports what the package achieved and is not an instance of the guarantee. The referee’s design choices were settled through ablation on synthetic streams before any paid run (Table 7, Appendix B).

7.2 One analysis, three environments

Every comparison is a paired difference between two arms that saw the same world. Within one model family and one start year, arms on the same seed share common random numbers: the same world and candidate draws. Their metric difference is one observation, and there are five seeds. The test is a two-sided paired t -test on those five differences. Holm correction covers the tests within one model family. Families are never pooled. An effect is reported as “ k of n families”. Appendix A states the conventions behind every table.

The synthetic world isolates the mechanism, because truth is planted. It has four families with true-factor proportions 0.50/0.25/0.10/0.05 and AR(1) noise with $\rho = 0.2$. Every admitted factor lowers the share of true factors among its family’s later candidates. One family has a mid-campaign regime break. A campaign lasts 1260 days, with four submissions every 21 days. The design is two LLM families $\times$ four referee arms $\times$ three controllers $\times$ five seeds.

The probe environment measures the one agent capability we claim: writing a diagnostic instrument for a fault nobody listed in advance. An episode shows the controller a deployed factor whose series has one hidden fault. The controller must choose an intervention. The score is *intervention regret*: the value of the best intervention less the value of the chosen one. The five fault types are held out, with twenty episodes each. They are blind spots of the three instruments in the fixed diagnosis menu. The probe-authoring arm may write its own instrument, test it in a sandbox, and read it before choosing. Seven model families are sampled from a written frame (Appendix D).

The real-data world tests which results survive ten years of prices. It is built point in time from CSI 500 constituents, 2015-01 to 2026-08 (Appendix E). The four-family library contains the technical families: momentum, short-horizon reversal, volatility and liquidity. The nine-family library adds value, size, quality, growth and investment. Campaigns start in 2016–2019, with five seeds each and four submissions every 21 days. A candidate’s realised label indicates whether its post-submission mean rank-IC is at least $\delta = 0.015$. Coverage is complete for the scripted controllers and partial for the LLM (Table 23, Appendix D). Three LLM families run the frozen referee on both libraries. One family runs the three leaky referees on the four-family library. The other two run its no-gate arm. Round-robin and bandit baselines run every arm at no cost. That makes 540 cells: 320 baseline and 220 LLM. Every LLM call is recorded, so every cell replays at zero cost (Proposition 2).

7.3 What is fixed before the data, and what is not

Every input is sorted by when it was fixed.

- **Fixed in advance and never tuned:** the level α , the viability threshold δ , the run-length target A , the cap fraction, the incubation of the no-gate arm, the parameter grids, and the design and analysis plan.
- **Walk-forward by construction:** every referee verdict, every controller action, and every sleeve decision

by the execution layer.

- **In sample:** the conversion constant $\bar{\kappa}$ was measured once over the whole panel. The execution layer was constructed after a first pass over the recorded cells (Appendix B).

There is no held-out period beyond the walk-forward, and there is one market. A second market needs data, not model calls. It is the natural next study.

7.4 The canonical book

The canonical book is the portfolio construction used for every reported cell. **Its numbers are in sample with respect to the execution layer** and make no out-of-sample claim. The referee-side results are computed before any book exists (§5.4). The construction has four steps.

- **Sleeves.** Each deployed factor forms a quintile long–short sleeve with one unit notional on its own calendar.
- **Calendar.** The execution layer selects each sleeve’s holding frequency from five desk frequencies. It scores them on the trailing 250 days with $c = 15$ bp and $\bar{\kappa} = 0.018$. A sleeve that cannot pay at any frequency is shelved.
- **Aggregation.** Traded sleeves are combined with equal notional. The book’s one-way turnover is capped at 25% per day. An index leg hedges its index beta.
- **Costs.** Costs are 15 bp per side on stocks and 5 bp on the hedge. They are charged on position changes.

Each book has three reported statistics. The CSI 500 index beta shows what the hedge leaves. A three-factor alpha follows the spirit of CH-3 (Liu et al. 2019) and uses Newey–West t -statistics (Newey and West 1987). The deflated Sharpe ratio (Bailey and López de Prado 2014) discounts for the 108 book-level groups in this study. Sharpe ratios use 244 trading days a year. Appendix A defines each statistic operationally.

8 Results

Three results carry the paper. The referee, not the controller, sets the number of false admissions (§8.1, §8.4, §8.5). The language model finds more true factors than a script and about as many as a bandit. Its clearest contribution is writing its own probes (§8.2, §8.6). The certified portfolio trails the ungated one because a daily certificate arrives late. It also trails because the family that pays least clears the certificate fastest (§8.7). Two controls (§8.3) rule out the two artefacts a reviewer would first suspect.

8.1 Synthetic factorial: the referee decides false admissions; the controller does not

Question. With truth planted, who determines the number of false factors admitted? *Answer.* The referee alone (Table 8, Appendix C).

False admissions follow the referee arm, regardless of the controller. Under the frozen referee, both LLM families admit 0.000 false factors per submission. Under the three leaky referees, the rate is 0.26–0.85 for script, bandit and LLM alike (2 of 2 families). The guarantee bounds the expected realised false-discovery proportion. This proportion is 0 in every seed under the frozen referee and 0.75–0.95 under the leaky ones. The guarantee does not reduce controller yield. Under the frozen referee, the three controllers have the same yield (0.095 round-robin, 0.096 bandit, 0.096 LLM, ± 0.002). Yield is lower under the leaky referees. Every admitted factor lowers the share of true factors among its family’s later candidates. A referee that admits

everything spoils the families it draws from. The no-gate arm’s yield is 0.042 against the frozen referee’s 0.095.

An attacking controller extracts false admissions only from a leaky referee. Hidden retries resubmit unadmitted candidates under new identifiers rather than drawing fresh ones. They extract +0.027 false admissions per submission from a peeking referee and +0.009 from an adaptive-threshold one, and none from the frozen one.

8.2 Probes: the agent as instrument-maker

Question. Is there a lifecycle task that the agent can do but a bandit cannot? *Answer.* It can write its own instruments. This has a clear effect in half of the evaluable model families and causes no harm in the rest (Table 9).

The comparison uses the fixed diagnosis menu because a bandit can only choose among listed actions. It cannot write a new one. Every arm sees the same episodes, so each episode provides a paired contrast. Authored probes lower intervention regret by 0.388, 0.248 and 0.233 in three families. Each result is significant after Holm correction over the six evaluable families. In the other three, the differences are -0.067 , -0.021 and $+0.074$ and cannot be distinguished from zero. One of the seven sampled families never produced a probe within its budget and is inevaluable.

8.3 Controls

Two artefacts could produce the LLM’s results without skill: what it remembers and randomness in its decoding. Neither does.

The memory control varies what the LLM may remember. Six memory arms (none, working, episodic, full, shuffled, oracle) run under the frozen referee. Each is compared with the full-memory arm. No difference is detected. After Holm correction, 0 of 10 tests are significant. Every arm is within ± 0.0025 yield of the full arm.

The decoding control tests whether rerunning the same model on the same history changes its decisions. A same-policy fork replays a recorded cell prefix, then lets the same model continue live. In 5 of 5 cells, the fork reproduces every outcome-level metric of the original.

8.4 Real data: the referee

Question. Does the synthetic result survive ten years of real prices? *Answer.* Yes. The referee still sets the number of realised false admissions. On real data, the size of this effect depends on who proposes (Table 2; Figure 6, Appendix C).

Table 2: Real data, four-family library: realised false admissions per campaign and realised yield per submission by referee arm and controller. Means cover four start years $\times$ five seeds; the LLM column is one model family.

Referee	False Adm.: Round-Robin	Bandit	LLM (Gemini)	Yield: Round-Robin / Bandit / LLM
Frozen	11.7	9.4	11.7	0.40 / 0.47 / 0.47
Peeking	86.2	38.4	37.4	0.41 / 0.50 / 0.55
Adaptive Threshold	105.2	51.0	43.5	0.39 / 0.44 / 0.48
No Gate	196.0	189.8	78.0	0.37 / 0.37 / 0.59

The proposer matters only when the referee leaks. Moving from round-robin to the LLM changes the count by a factor of 2–2.5 on the leaky arms. It does not change the count on the frozen one. The bandit halves the count under peeking and adaptive threshold. It leaves the count unchanged under no gate.

The two kinds of referee also admit different kinds of false factor. This difference matters for the book. A false admission with a negative realised edge loses money for as long as it is held. One with a positive edge below δ merely fails to pay for its trading. The frozen referee’s realised false admissions are all of the second kind. None of its 100 four-family cells has an admitted factor with a negative realised edge. The detector had retired every false admission before the campaign ended. Under round-robin, the leaky referees admit 10.8, 32.0 and 90.8 negative-edge factors per campaign. Under the LLM, they admit 1.2, 6.0 and 15.5 (Table 10).

The ordering survives every label we tried. The size of the ratio depends on the label. Moving the threshold from 0.010 to 0.020 changes every count. It does not change the ordering (Table 11). Judging each admitted factor only on post-admission days removes the overlap of §3. On that label, the frozen referee’s false admissions under round-robin rise from 11.7 to 17.6 per campaign. The leaky referees’ false admissions barely move. Its advantage is then $5\text{--}11\times$ against $7\text{--}17\times$ on the registered label. Under the LLM, it is $2.6\text{--}4.9\times$ against $3\text{--}7\times$ (Table 12).

Against each family’s own break-even, the ratio shrinks to two- to threefold. Relabelled that way, the frozen referee has 96.0 false admissions per campaign under round-robin. The three leaky referees have 180.4, 197.4 and 276.2. These counts arise because short-horizon reversal has a break-even of 0.074 (§4.2). The headline ratio is therefore a statement about the registered label. The frozen referee certifies a statistical edge. Whether that edge pays is the execution layer’s question (§8.7).

8.5 Real data: a legitimate comparator

Question. The three leaky referees are known shortcuts. What does the anytime-valid referee add over a patient desk that waits and corrects for multiplicity? *Answer.* For clearly good or bad families, it adds a little accuracy and some yield. For many families with a small positive edge, it gives a fourfold gain in accuracy. That gain comes from the height of its bar (Table 14, Appendix C).

The comparator is the procedure a careful desk would run without anytime-valid tools. It tests each candidate once, 500 trading days after submission (the frozen referee’s median wait). It uses a one-sided t -test on the whitened stream. It applies Benjamini–Hochberg at 0.05 every time a window closes. It runs as a counterfactual program on the frozen cell’s recorded submission stream.

On the four-family library, the patient desk comes close, at some cost in yield. Under round-robin, the 500-day comparator admits 16.4 false factors per campaign against the frozen referee’s 11.7. Both are far from the 86–196 of the leaky referees. Its yield is lower, at 0.358 against 0.403. Its wait is a fixed 500 days against a median of 439. The frozen referee lets strong factors through early.

On the nine-family library, the comparator is four times less accurate because its bar is lower. It admits 41.4 false factors against 9.8. After the decision, 0.33 of what it admits is false, against 0.12. Both procedures test a zero edge, and both are valid for it. A t -test on 500 days has power against an edge of 0.01. This edge is positive and below the viability threshold. The fundamental families supply many such factors. The frozen referee’s bar of $N_v/(k\alpha)$ is far higher. Therefore, only strong factors clear it within a campaign. Its selectivity and its wait are the same knob (§4.3).

The evidence therefore supports the claim that the judge must be a valid, multiplicity-corrected statistical procedure. Either valid procedure admits several times fewer sub-threshold factors than the shortcuts. Anytime

validity then adds three things. The verdict can be read on any day without choosing a horizon in advance. Strong factors pass early. The guarantee also holds for a proposer that adapts to the verdicts. The comparison has one limit. The candidate sequence is the one the frozen referee produced. Therefore, the comparison isolates the referee and is not a re-run campaign.

8.6 Real data: the controller

Question. What does a language model add over a script or a bandit, given a valid referee? *Answer.* It finds more true factors than a script and about as many as a bandit (Table 3). It admits fewer false factors than either only when the referee is broken (Table 10).

Here is how to read Table 3. Each entry contains four signs, one for each start year. Each sign shows the paired difference between the LLM controller and a baseline. The number in parentheses counts the starts where that difference is significant after Holm correction. Tables 21 and 22, Appendix C, report paired effects with standard errors.

Table 3: Real data, frozen referee: signs of the paired difference between the LLM controller and baseline across the four start years, by model family. Parentheses show the number of starts significant after Holm correction within the family. Families are never pooled.

Library	Contrast	Metric	DeepSeek	Gemini	GPT
Four-Family	LLM – Round-Robin	Yield	+ + + + (4/4)	+ + + + (3/4)	+ + + + (2/4)
Four-Family	LLM – Bandit	Yield	+ + – –	+ – – –	– – – –
Nine-Family	LLM – Round-Robin	Yield	+ + + + (4/4)	+ + + + (3/4)	+ + + + (1/4)
Nine-Family	LLM – Bandit	Yield	+ + + –	+ + + –	– – – – (3/4)
Four-Family	LLM – Round-Robin	False Adm.	+ + + – (0/4)	– + + + (0/4)	+ + + + (0/4)
Nine-Family	LLM – Round-Robin	False Adm.	+ + + +	+ + + +	+ + + +

The LLM finds more true factors than round-robin and no fewer false ones. Its realised yield is higher in six of six family $\times$ library settings. The difference is positive in every start year. Its false admissions under the frozen referee are at or slightly above round-robin’s (11.7 to 13.3 against 11.7). This occurs because it sends more submissions to productive families. The increase is Holm-significant in no start for any family.

The balance against the bandit is level. It depends on the library. On the four-family library, the LLM is level or behind in most starts. On the nine-family library, two families lead in three of four starts. One loses in every start (three of four significant). That library has five barren families, which punish an allocator that keeps sampling them.

Under leaky referees, the LLM’s judgement replaces part of the missing referee. It reduces realised false admissions relative to round-robin on all three leaky arms (Holm 4/4 each; three model families of three on the no-gate arm). Relative to the bandit, it does so only under no gate (4/4). There, the bandit’s admission reward stops discriminating, while the LLM still reads profiles and retirements.

A period split suggests that the LLM’s advantage over the bandit is a prior that data overtake. We split every campaign at 2020-01-02. This is the first trading day of the period when the fundamental families lose their edge in this market. Against the bandit, the LLM leads before the split and trails afterward in all three model

families (Table 17, Appendix C). Early in a campaign, the bandit has no admissions to learn from. The LLM’s family profiles are therefore informative. Later, the bandit has learned what the profiles said. This is an interpretation, not a test. It does bear on leakage. A model that had seen the period would be expected to gain most after 2020. The sign is the other way.

8.7 Real data: the portfolio

Question. What does a portfolio of certified factors earn, and what does the certificate cost an investor?

Answer. The certified book does not lose money at the registered cost. Its alpha cannot be distinguished from zero. Its Sharpe ratio net of costs is below that of a book that admits everything. The gap has two causes: a horizon mismatch and the wait. Neither is a verdict on the certificate.

The certified book earns money net of the registered costs, but not demonstrably. The four-family library has twenty frozen-referee groups of five seeds. Their net Sharpe is +0.33 to +0.50. Against the study’s 108 groups, the deflated Sharpe ratio is 0.49–0.69. This is far from the level near one that survives a search of that size. Alpha is +0.25 to +1.13 bp per day. Its Newey–West t never reaches one in any group. Index beta is -0.07 to -0.08 . The nine-family library gives the same picture.

The ungated book ranks first in every start year. Its net Sharpe is +0.59 to +0.77, versus the frozen referee’s +0.33 to +0.50. The two leaky referees fall between them. The order survives when every book is restricted to 2020 onwards. It also survives re-costing (Appendix B).

The first cause of the gap is a horizon mismatch. The ungated book holds momentum, which a daily certificate cannot see. The per-candidate ledger records each factor’s own sleeve returns for every day it is held. Thus, the two books can be compared line by line (Table 4). The two books earn about the same per day from volatility and liquidity. Momentum creates the large difference. The ungated book held 892 momentum sleeves at +1.69 bp per day. The frozen book held six.

The referee is right about momentum on its own terms. It is wrong for the book. Momentum’s daily rank-IC is 0.001, an order of magnitude below δ . It therefore cannot be certified within a campaign. But momentum is a slow factor in this market. Its information coefficient *rises* with the holding day, from 0.001 at one day to 0.009 at sixty-three. It pays only when held a month or longer (Figure 4). A library that certifies one-day predictive power cannot certify momentum. A library that certifies nothing includes it incidentally.

Table 4: Per-family sleeve economics under the canonical book, using the round-robin controller and four-family library. Results pool five seeds from the 2016 start. They report sleeves ever held, the share shelved, and each family’s daily earnings while held, **gross of their own trading costs**. Re-formed daily, a reversal sleeve nets -19.6% per year (Figure 4).

Family	Frozen: Held / Shelved / bp per Day	No Gate: Held / Shelved / bp per Day
Volatility	473 / 1% / +4.05	418 / 2% / +4.06
Liquidity	314 / 0% / +3.79	555 / 17% / +3.96
Short-Horizon Reversal	539 / 30% / +2.26	475 / 41% / +1.50
Momentum	6 / 83% / -4.35	892 / 44% / +1.69

The second cause is the wait. The wait counts the trading days between a factor’s submission and admission. For an admitted true factor, its median is about 500 trading days under the frozen referee. It is 122 to 214 under the leaky referees (Figure 7, Appendix C). Figure 5 shows the wait’s effect on a book. The ungated book begins compounding in its first year while the certified book remains empty. Once both are invested, the two slopes are similar. The gap opens during the wait and is not earned afterwards.

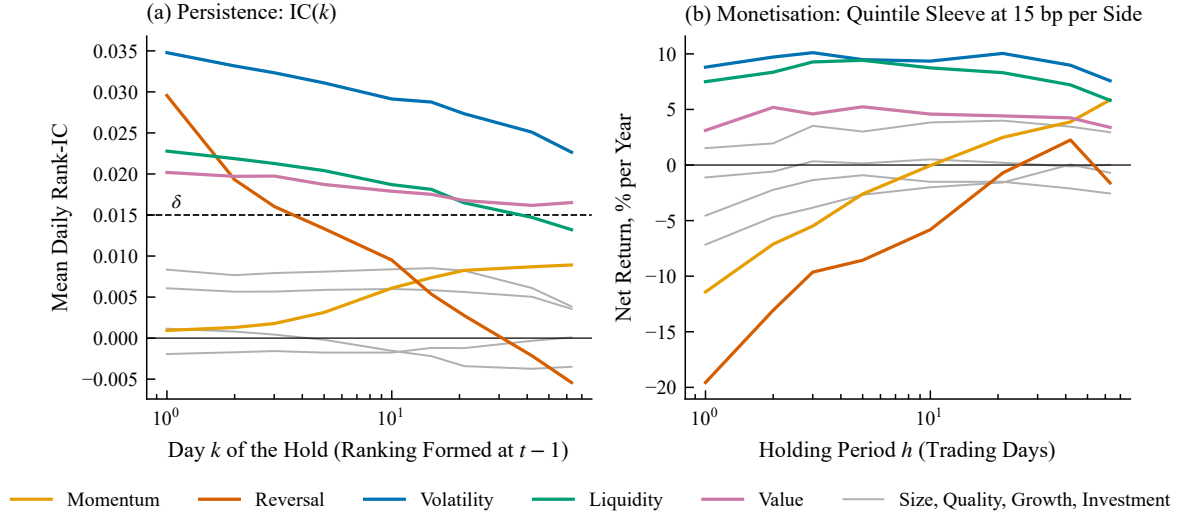


Figure 4: Persistence and monetisation by family. Left: the decay curve $IC(k)$ of a ranking known at $t - 1$ against the return on day $t + k - 1$, with the viability threshold δ . Right: what a single-factor quintile sleeve nets per year at 15 bp per side when re-formed every h trading days. Reversal nets at most +0.9 bp per day at any frequency on the grid; momentum is not certifiable at one day and pays only when held a month or longer.

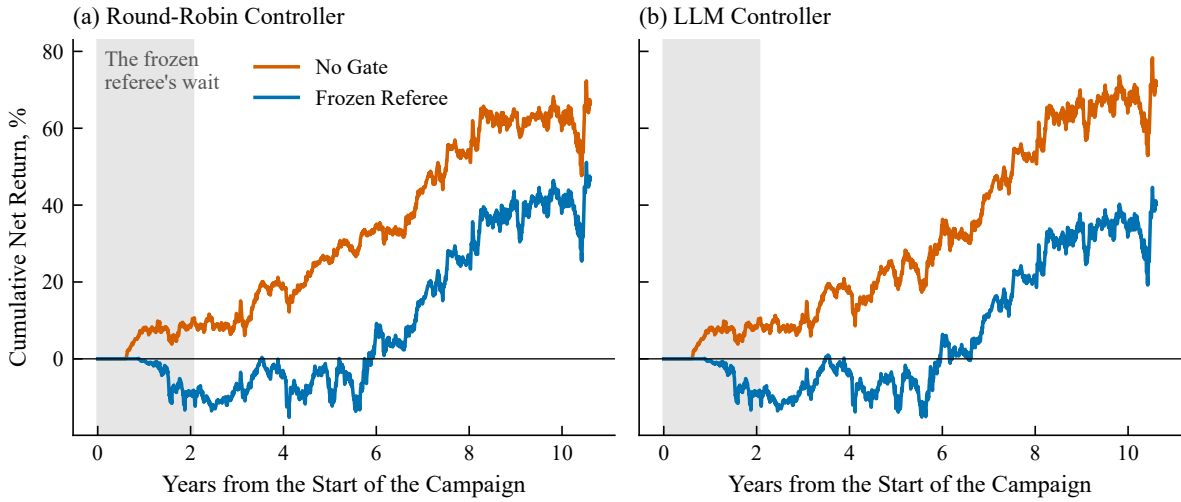


Figure 5: Cumulative net return of the canonical book, 2016 start, seed-averaged (four-family library; left the round-robin controller, right the LLM controller). The shaded band is the frozen referee's wait. The ungated book begins compounding in its first year while the certified book is still empty, and is below water until year six. The two slopes are similar once both are invested.

The wait is expensive because it is paid in regimes, not days. In the nine-family library, 99–100% of certified holding windows are centred after 2020-01-02. The corresponding range is 65–100% by family for the ungated pipeline (Table 20, Appendix C). In this market, the edges of the fundamental families value, quality and growth disappeared after 2020. The certified book earns nothing from exactly those families. Waiting for volatility and liquidity costs almost nothing because their edge is stable. A slow certificate is cheap when the edge persists and expensive when it decays. A descriptive decomposition of the Sharpe gap agrees. Earliness

is its largest term on the three longer campaigns (Table 19).

Two counterfactual books test this account (Table 16, Appendix C). Both use the round-robin frozen cell’s own candidate sequence and the world’s own accounting. The frozen book is reproduced day by day in 20 of 20 cells.

The first counterfactual switches retirement off. It shows that retirement is not what the certified book loses. Every admitted factor is held until the campaign ends. Net Sharpe changes by +0.00 to +0.01 across the four start years. Some real-data retirements are false on the realised label. Reversal factors were retired after about 838 days. Their edge had sagged to 0.0145 while held. It averaged 0.034 over the rest of the campaign (Table 18). Those false retirements cost the book almost nothing.

The second counterfactual replaces the frozen referee with the 500-day comparator. It shows that a desk with the same patience builds a better book without being earlier. The comparator’s net Sharpe is +0.59 to +0.65, against +0.40 to +0.49 for the frozen book. This is a paired gain of +0.12 to +0.24 in every start year. Its factors are not admitted sooner. The mean wait is 500 days against 510. The mix differs. The comparator admits fewer reversal factors, 72.0 against 87.2. It admits about five momentum factors, while the frozen referee admits almost none, 4.8 against 0.5. The highest daily rank-IC reaches the bar fastest, so the bar favours the family with the worst net economics. A referee with the same patience does not pay that cost.

Certification still buys one thing the Sharpe ratio does not show. The execution layer never sees the referee. It shelves 13% of what the frozen referee admitted, against 26% of what the ungated pipeline deployed. More of what the referee admits is worth trading. A portfolio Sharpe ratio is therefore not a verdict on certification in either direction. It combines a referee with an execution layer that is blind to it. The same holds for LLM investment agents evaluated as a whole: the capability leader is not the Sharpe leader, and the process must be diagnosed rather than a period’s return ranked (Qu and Chen 2026).

9 Discussion and limitations

The experiments support the division of labour and show its cost. The judge must be a statistical procedure the agent cannot touch. Every leaky referee admitted several times more false factors than the frozen one under every proposer. No proposer closed that gap. A fixed-horizon test with a multiplicity correction is also a statistical judge. Anytime validity adds a verdict that can be read on any day. Its selectivity is paid for in waiting.

The proposer can be anything. A language model earns its place through jobs a bandit cannot do. It does not allocate consistently better than a bandit. It adds three things. It reads why a factor was retired. It writes an instrument for a fault the menu does not cover. It also holds the line when the referee leaks.

The split costs time. This time reflects an information bound rather than an engineering gap. An admitted true factor waits a median of about five hundred trading days in this market. The wait scales like $\ln(N_v/(k\alpha)) \cdot 2\sigma^2/\mu^2$. At this noise level, no bettor can certify a factor at the threshold in under five years. Only a less noisy statistic or a lower bar shortens the wait.

The comparator shows a second price. A desk with the same patience and a fixed-horizon test builds a better book. Its admissions are less concentrated in short-horizon reversal. An anytime-valid bar rewards the statistic that is strongest per day. The strongest daily statistic in this library is the hardest to trade. The design change is to certify the horizon that is traded, not the daily one.

The evidence has clear edges, and each edge bounds one claim:

- *The sample.* The sample covers one market and four start years that share most of their data. There is no held-out period beyond the walk-forward.
- *The label.* The registered label leans towards the frozen referee. It is not planted truth and overlaps the admission window. Judging each factor only after its admission lowers the headline ratio by about a third. The synthetic factorial has no overlap and carries the claim about false admissions.
- *The comparisons.* The LLM runs the leaky referees only on the four-family library. The fixed-horizon comparator is a counterfactual referee on the frozen referee’s candidate sequence. It is not an arm with its own campaign. The diagnosis role of the LLM controller is not ablated on its own. The controller contrasts compare packages, since the LLM also reads profiles and memory, and on real data they are replay outcomes, not forecasts.
- *The guarantee.* The proof controls the conditional null on the whitened stream. The marginal-null guarantee under autocorrelation is measured, not proved. The cluster merge is not exercised.
- *Retirement.* Retirement is calibrated on synthetic decay. Its guarantee concerns the average run length. This gives no usable bound over one campaign.
- *The portfolio.* The portfolio results support no claim of investability. Alpha t -statistics are below one in every certified four-family group. The cost model includes no borrow fee. The book is an instrument, not a strategy (Appendix B).

10 Conclusion

Judging must be governed by a statistical procedure that the agent cannot touch. Proposing and instrument-making are where the agent’s intelligence has measurable, bounded value. This division of labour is governed self-evolution, the framework we have proposed and measured. The agent’s surface evolves freely. A frozen referee decides what enters the portfolio. Who judges determines how many false factors an always-on miner certifies. No proposer substitutes for the judge.

The split is testable because three objects remain separate and auditable: the certificate for the statistic, the measurement of its persistence, and the execution decision. The certificate is judged on the referee’s own daily stream. Persistence and execution are measured afterwards, on returns the certificate never saw. A portfolio Sharpe ratio mixes all three. That is why it is evidence neither for nor against a certificate. It also explains the next design question: the certified horizon should match the traded one.

Data and code availability

The code, run manifests, prompts, and recorded model calls are available from the corresponding author on request (deepgroundingai@gmail.com). The authors cannot redistribute the market data because they license them from a commercial data vendor.

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Appendix A Definitions, conventions and worked calculations

This appendix supports the body on three things. It defines the terms the body names before it builds them. It states the conventions behind every reported number. It works three calculations the body quotes. Nothing here is a new result.

Terms at first use. An *e-process* is a candidate's betting capital read over time. Under the null it is a test supermartingale, so its value at any stopping time is a fair quantity, an *e-value* (§4.1). *Online e-BH* is the Benjamini–Hochberg procedure applied to those e-values as they arrive, over a universe of hypotheses fixed in advance (§4.3). An *e-detector* is a sum of e-processes restarted every day, used as a changepoint chart (§4.4). A *leaky referee* is a referee that breaks one assumption of Proposition 1 in the way a common practice does (§7.1). *Planted truth* means that the synthetic world knows which of its factors have an edge, so every admission is labelled true or false without a proxy (§7.2). A *perception probe* is a small program the controller writes to measure a suspected fault in a deployed factor. Its output informs the controller's diagnosis and never enters a statistical test (§6). A *sleeve* is one factor's own quintile long–short portfolio. A sleeve is *shelved* when it cannot cover its costs at any holding frequency (§5).

Candidate clocks. A candidate is open, deployed or retired (§3). A candidate that leaves the open state without admission has *expired*. It keeps its slot (§4.3). A *false retirement* is an alarm of the retirement detector raised while the deployed factor's edge is still at least δ . The *run length* is the number of trading days from the detector's start to its alarm. The guarantee of §4.4 bounds the expected run length to a false retirement from below by $A^* = 1260$ days. Over one campaign that bound is not informative. The false-retirement rates reported are therefore simulation results.

Admission bookkeeping, with a two-candidate example. An *epoch* is a period in which the library is frozen. The reported runs use one epoch. A *slot* is one of the $N_v = 2000$ places in the registered universe. Every submitted candidate takes one slot and keeps it, whether it is admitted, expires or is retired. With $w_v = 1$ every slot has weight $\gamma_c = 1/N_v$. The bar for the k -th admission at level $\alpha = 0.05$ is $N_v/(k\alpha)$. It is 40,000 for the first admission and 20,000 for the second. Suppose candidate A has capital 45,000 and candidate B has 25,000. A alone clears the first bar. Both clear the second bar, so $k^* = 2$ and both are admitted. If B had 15,000 instead, only A would be admitted and B would keep betting. The rule lowers the bar as more candidates clear it. It never raises the bar.

The wait, worked. At the Kelly stake the expected wait for the k -th admission is $T \approx \ln(N_v/(k\alpha)) \cdot 2\sigma^2/\mu^2$ (§4.3). For the first admission $\ln(40,000) \approx 10.6$. Daily rank-IC has $\sigma \approx 0.12$. At edge $\mu = 0.03$, $2\sigma^2/\mu^2 = 32$, so $T \approx 339$ days. At the viability threshold $\mu = 0.015$, $2\sigma^2/\mu^2 = 128$, so $T \approx 1356$ days. The capped bettor as run, with the stake at the cap of 0.8, takes 546 and 1434 days in the same two cases. A universe sized to the campaign's 490 slots would lower every bar by $\ln(2000/490)$. That is worth 72 days at edge 0.03 and 191 days at 0.015.

The four threats and their answers. The body names four threats (§3). Each has one answer.

1. **Optional stopping.** The capital is a test supermartingale. It may therefore be read every day and stopped by any rule based on the past, and the false-discovery bound holds at the stopping time (§4.1).
2. **Near-copy resubmission.** Every copy takes its own slot under online e-BH. The bound stays at α . Repetition spends the proposer's slots and nothing else (§4.3).
3. **Verifier-in-the-loop adaptation.** The referee sizes each bet before the day's outcome and scores only data that arrived after submission. What the controller learns about the referee changes which candidates it submits, not the referee's error (Proposition 1).

4. **Knowledge-cutoff leakage.** The referee does not address it. Proposition 1 assumes that the proposer holds no information about the post-submission stream. A live run satisfies this. A historical replay by a model trained on the campaign years may not. The language-model arms’ real-data results are therefore *replay outcomes*: what the package achieved on that history, not an instance of the guarantee.

The analysis unit and the reporting conventions. The unit of analysis is a seed. Two arms on the same seed, model family and start year share common random numbers, and their metric difference is one observation. A reported “ $\pm$ ” is the standard error over the five seeds. The test is a two-sided paired t -test on the five differences. Holm correction is applied within one model family. On real data that family has 24 tests: four starts $\times$ two contrasts $\times$ three metrics. In the synthetic world it has 8: four referees $\times$ two metrics. An effect is reported as “ k of n families”. In the sign tables, each entry has one sign per start year. The number in parentheses counts the starts significant after Holm correction. No parentheses means none.

Portfolio statistics, operationally. A *group* is the five seeds of one referee, controller, library and start year. The study has 108 groups. The net Sharpe ratio is the mean daily net return over its standard deviation, annualised with 244 trading days. The deflated Sharpe ratio (Bailey and López de Prado 2014) discounts that ratio for the 108 groups compared. Alpha is the intercept of a regression of daily net return on market, size and book-to-price legs built from the panel, in the spirit of CH-3 (Liu et al. 2019), with Newey–West t -statistics at lag 5 (Newey and West 1987). It is quoted in basis points per day. Index beta is the slope on the CSI 500 index return.

The factorial in one schema. Each environment has a unit, an outcome and a contrast.

- **Synthetic world.** Unit: a cell (one arm combination, one model family, one seed). Outcomes: realised yield per submission and realised false admissions per campaign, labelled against planted truth. Contrasts: four referee arms $\times$ three controllers, two LLM families, five seeds.
- **Probe environment.** Unit: an episode (one deployed factor with one hidden fault). Outcome: intervention regret. Contrast: the authored probe against the fixed diagnosis menu on the same 100 episodes, per model family, seven families.
- **Real data.** Unit: a cell. Outcomes: the realised label at $\delta = 0.015$ and the canonical book’s statistics. Contrasts: the same arms, four start years $\times$ five seeds, with LLM coverage as in Table 23.

Appendix B Derivations and ablation details

Bettor ablation. The bettor choice is immaterial at these noise levels. The cap fraction sets the wait. At $\mu = 0.03$ and $\sigma = 0.12$, the Kelly stake exceeds any no-bankruptcy cap. Thus, aGRAPA, the Gaussian-efficient bettor (Martinez-Taboada and Ramdas 2026) and a capped oracle all sit at the cap. A cap fraction of 0.8 accepts a $\mu = 0.03$ factor in a median of 162 days ($\rho = 0$) or 190 days ($\rho = 0.2$). At a fraction of 0.5, the median is 238/284, with null false acceptance of 1.9/3.4%.

Holding grid. The grid covers the five frequencies a desk trades: weekly, fortnightly, monthly, bi-monthly and quarterly. It is scored using the average of the decay curve. It does not use the practitioner’s “IC at horizon h ”. That measure rises mechanically with h because averaging returns lowers noise. The design excludes daily formation. This costs under 1% per year for the two families flat enough to favour it.

Conversion estimator. A conversion rate estimated per sleeve is too noisy to guide shelving. The first canonical pass estimated it per sleeve. This reversed the cost curve. It shelved 77% of volatility and 48% of liquidity sleeves, against 32% of reversal. The denominator caused this reversal. It is a 250-day mean IC whose standard error (0.006–0.013) matches the size of a real edge. With the market constant, the cells shelve 0% of

liquidity, 2% of volatility, 30% of reversal and 89% of momentum.

Cool-down sweep. The pre-stated rule *never worst, least churn* selected the cool-down $B = 250$, not Sharpe. Dynamic variants with $B \in \{5, 10, 21, 63, 250\}$ differ by 0.4–3.8% per year by family. No variant wins consistently. $B = 250$ exceeds fixed monthly in four of five families. It is the only variant that turns reversal positive.

The conversion constant is in sample. $\bar{\kappa} = 0.018$ is the one input measured over the whole sample. Across the half-sample band 0.013–0.026, the traded-or-shelved decision flips for 3.3–7.2% of the 51,531 calibrated sleeves. Almost all flips are in reversal (14–19% of its sleeves). The books were not re-booked across the band.

Cost sensitivity. Table 5 re-costs the 2016 round-robin books with fixed positions and scaled stock-side cost. The certified book’s one-way turnover is 3.4% per day, against the ungated book’s 2.2%. Thus, “the certified book does not lose money” applies at the registered cost level.

Table 5: Cost sensitivity of the two books (five-seed means). The certified book crosses zero between 30 and 50 bp a side. The ungated one does not.

Cost per Side	Certified Net SR	Ungated Net SR
15 bp (registered)	+0.40	+0.73
25 bp	+0.25	+0.61
30 bp	+0.17	+0.54
50 bp	−0.12	+0.29

Accounting conventions. Reviewers questioned two conventions of the registered book. Neither changes the comparison (Table 6). *Carry* holds a halted position at its last price and credits the whole gap on the reopening day. It does not book a missing return as zero. *Lag* executes the target book decided on day t on day $t + 1$. It does not execute at the signal’s own close. Both are re-booked at zero cost from the recorded holding intervals. The canonical re-booking reproduces every recorded book to 10^{-9} .

Table 6: The 2016 round-robin books under four accounting conventions (five-seed means with standard errors, 244-day annualisation). The ungated book’s lead is 0.34–0.35 under every convention.

Book	Convention	Net Sharpe	Cumulative Net Return
Certified	Registered	+0.40 ± 0.03	+46.6%
Certified	Carry	+0.40 ± 0.03	+47.1%
Certified	One-Day Lag	+0.37 ± 0.03	+43.5%
Certified	Carry and Lag	+0.38 ± 0.03	+44.1%
Ungated	Registered	+0.73 ± 0.02	+66.3%
Ungated	Carry	+0.75 ± 0.02	+67.7%
Ungated	One-Day Lag	+0.71 ± 0.01	+64.1%
Ungated	Carry and Lag	+0.72 ± 0.01	+65.5%

Instrument properties. The book is a measuring instrument. The paper gives no certificate of tradability. Short selling is assumed for a quintile of individual A-shares, without a borrow list or borrow fee. The hedge leg uses index futures. A missing return is booked as zero (12.5% of member-stock days in 2015, about 0.2% from 2019). Neither capacity nor drawdown is modelled. Measured maximum drawdown is 21–25% for the certified book and 20–21% for the ungated one.

Referee ablations. The adopted referee whitens the stream. It uses the predictable bankruptcy bound and estimates ρ . Ablations on synthetic streams settled each choice before any paid run (Table 7). Whitening restores nominal control at every autocorrelation tested. The predictable bound recovers most of the power

lost under a worst-case bound. Estimating ρ beats knowing it at $\rho = 0.4$. This advantage arises because the shrunk estimate under-whitens, trading a within-bound validity error for power.

Table 7: Referee ablations use synthetic AR(1) streams at $\rho = 0 / 0.2 / 0.4$. This table uses cap fraction 0.5. The referee runs at 0.8. The experiment uses ten-year null streams at $\mu = 0$ and alternative streams at $\mu = 0.03$. Alive streams cover 2520 days with run-length target 1260. The nominal level is 5%. Against a Monte-Carlo-calibrated CUSUM, the adopted e-detector’s median delay is 204–322 days by decay shape, against CUSUM’s 181–337.

Stream Transformation	Null False Acceptance, %	Median Days to Accept	Detector False Alarms on Alive Streams, %	Step-Decay Delay, Median Days
Raw daily rank-IC (conditional null)	2.1 / 6.0 / 15.4	228 / 227 / 227	1.5 / 11.5 / 46	213 / 181 / 195
Whitened, worst-case bound	1.1 / 1.3 / 1.8	336 / 378 / 462	1.0 / 0.0 / 7.0	250 / 264 / 316
Whitened, predictable bound (adopted)	1.8 / 2.0 / 2.4	238 / 284 / 384	1.5 / 0.5 / 8.5	218 / 234 / 308
Whitened with the true ρ (oracle)	2.1 / 1.3 / 0.9	228 / 344 / 536	1.5 / 0.5 / 6.5	213 / 246 / 333

Family removal. Removing each family in turn from the ungated book shows that momentum helps through diversification, not return. Dropping momentum costs +0.11 to +0.18 net Sharpe while raising cumulative net return in all four starts. A momentum-only book earns +0.22 to +0.29. Dropping the certified reversal family raises the book’s Sharpe by +0.01 to +0.14.

Selection. We chose the execution layer after first examining the recorded cells. One fact limits how much this choice can explain. Every construction tried (A10b, A11, A11b, A11c in Table 24) ranks the ungated book above the certified one. Neither execution parameter was chosen on Sharpe. The deflated Sharpe ratio treats the 108 final book groups, not the constructions, as trials. The reported deflation therefore understates the search.

Newey–West bandwidth. At Bartlett bandwidths 5, 8, 21 and 63, the certified book’s alpha t -statistics are 0.49, 0.49, 0.51 and 0.54. Those for the ungated book are 1.68, 1.71, 1.75 and 1.86. The inference therefore does not depend on the bandwidth. We keep lag 5.

Benchmark construction. The regression uses a characteristic three-factor model on our own panel, not the published CH-3 benchmark (Liu et al. 2019). SMB and HML are long–short quintiles of $-\log$ circulating capitalisation and book-to-price over the full universe. The benchmark legs are gross, while the dependent variable is net. This difference is expected to lower the estimated alpha.

Rank-threshold retirement. Declaring the top k streams at A^*/k satisfies the error-over-patience bound but is unusable. A healthy factor’s statistic plateaus near $1/(1 - \text{decay})$, with median ≈ 75 on the whitened stream. Yet A^*/k is 63–126 for $k = 10$ –20. The rule therefore declares an all-alive library of 20 en bloc within a year.

Appendix C Extended results

This appendix presents detailed evidence for §8. Tables 8 and 9 present the synthetic factorial and probe environment. Figure 6 plots Table 2. Tables 10 to 18 provide zero-cost ledger analyses for the real-data sections. They cover the realised edges of admitted factors, label-threshold sensitivity, and after-admission and per-family labels. They also cover legitimate comparators, counterfactual books, the period split, and retirement outcomes. Table 19 gives the Sharpe decomposition. Figure 7 shows the admission-delay distribution, and Table 20 reports the regime timing. Tables 21 and 22 provide the paired controller contrasts behind Table 3.

Table 8: Synthetic world with planted truth (four families, 1260 days, five seeds). The table reports seed-mean realised yield and false admissions per submission by referee arm and controller for the two LLM families run there. It also reports the hidden-retry attacker’s false admissions.

Family	Referee	Yield: Round-Robin / Bandit / LLM	False Admissions:	
			Round-Robin / Bandit / LLM	Hidden Retries: False
Gemini	Frozen	0.095 / 0.096 / 0.096	0.000 / 0.000 / 0.000	0.000
Gemini	Peeking	0.060 / 0.065 / 0.063	0.314 / 0.300 / 0.315	0.341
Gemini	Adaptive	0.081 / 0.087 / 0.087	0.272 / 0.273 / 0.263	0.281
	Threshold			
Gemini	No Gate	0.042 / 0.034 / 0.043	0.838 / 0.848 / 0.836	0.820
GPT	Frozen	0.095 / 0.096 / 0.096	0.000 / 0.000 / 0.000	0.000
GPT	Peeking	0.060 / 0.065 / 0.061	0.314 / 0.300 / 0.318	0.341
GPT	Adaptive	0.081 / 0.087 / 0.088	0.272 / 0.273 / 0.271	0.281
	Threshold			
GPT	No Gate	0.042 / 0.034 / 0.041	0.838 / 0.848 / 0.839	0.820

Table 9: Probe environment at the registered escalation price. For each model family, the table reports mean intervention regret against the oracle (edge units) $\pm$ s.e. over 100 episodes for the fixed diagnosis menu and authored probes. Paired differences are authored minus menu within each episode. Holm correction covers the six evaluable families.

Family	Fixed Menu: Regret	Authored Probes:		
		Regret	Paired Difference	Holm p
DeepSeek	0.616 $\pm$ 0.098	0.228 $\pm$ 0.064	−0.388 $\pm$ 0.101	0.001
Muse	0.651 $\pm$ 0.084	0.403 $\pm$ 0.064	−0.248 $\pm$ 0.070	0.002
Gemini	0.692 $\pm$ 0.098	0.458 $\pm$ 0.117	−0.233 $\pm$ 0.085	0.025
GPT	0.526 $\pm$ 0.078	0.459 $\pm$ 0.097	−0.067 $\pm$ 0.072	1.000
Hunyuan	0.663 $\pm$ 0.097	0.642 $\pm$ 0.109	−0.021 $\pm$ 0.069	1.000
GLM	0.585 $\pm$ 0.093	0.659 $\pm$ 0.148	+0.074 $\pm$ 0.114	1.000
Qwen	0.415 $\pm$ 0.103	0.415 $\pm$ 0.103	Inevaluable (No Probe Produced)	

Appendix D Environment, amendments and information parity

The 540 cells comprise 320 baseline cells and 220 LLM cells (Table 23). Table 24 lists every dated amendment to the pre-registered design.

The probe environment. The five held-out fault types include collapsed cross-sectional dispersion, possible crowding or regime-change decay, and lagged restatement of fundamentals. They also include sibling-factor contamination and a cost-channel fault. The fixed menu contains three instruments: a data-health canary, a crowding proxy and an IC changepoint test. The controller chooses one of five interventions: retire the factor, deprecate or pause its family, repair the data pipeline, or pay to escalate. A transparent economic model gives each intervention’s after-cost value in each true state over 126 days. Regret is the oracle’s value less the chosen intervention’s value.

Information parity. All controllers receive the same numeric family statistics. The LLM controller also receives family profiles, open candidates, paused families and memory. The controller contrasts therefore compare packages. No controller sees the book. The text names model versions only by family, while the run manifests pin them.

The break-even figure in the family profiles. The family profiles contain a scaling error that does not change how the controller contrasts are read. Their break-even ICs used the regression slope of \$5.2, not the average conversion. Thus, every figure is between a quarter and a third of the corrected value. The error is a common

Table 10: Anatomy of admissions on the four-family library (means per campaign over four starts $\times$ five seeds). The table splits ever-admitted factors by their realised post-submission mean rank-IC. The first two classes together are realised false admissions. The last column counts false admissions retired by the detector before campaign end.

Arm	Admitted	Edge < 0	$0 \leq \text{Edge} < \delta$	Edge $\geq \delta$	False Admissions	Retired by the End
Frozen, Round-Robin	214.7	0.0	11.7	203.0	11.7	11.7
Frozen, Bandit	260.8	0.0	9.4	251.3	9.4	9.4
Frozen, LLM (Gemini)	257.0	0.0	11.7	245.3	11.7	11.7
Frozen, LLM (DeepSeek)	258.1	0.0	12.7	245.5	12.7	12.7
Frozen, LLM (GPT)	236.9	0.0	13.3	223.6	13.3	13.3
Peeking, Round-Robin	312.5	10.8	75.3	226.3	86.2	85.6
Peeking, Bandit	342.8	3.6	34.7	304.4	38.4	38.1
Peeking, LLM (Gemini)	354.9	1.2	36.1	317.6	37.4	37.0
Adaptive Threshold, Round-Robin	322.3	32.0	73.2	217.1	105.2	93.7
Adaptive Threshold, Bandit	327.6	11.6	39.4	276.6	51.0	47.7
Adaptive Threshold, LLM (Gemini)	330.6	6.0	37.5	287.1	43.5	42.8
No Gate, Round-Robin	398.0	90.8	105.2	202.0	196.0	150.8
No Gate, Bandit	398.0	92.0	97.8	208.2	189.8	148.1
No Gate, LLM (Gemini)	398.0	15.5	62.5	320.1	78.0	61.8
No Gate, LLM (DeepSeek)	398.0	49.5	77.0	271.4	126.6	97.5
No Gate, LLM (GPT)	398.0	74.2	94.9	228.9	169.1	126.6

Table 11: Label-threshold sensitivity on the four-family library (means over four starts $\times$ five seeds). The table reports realised yield per submission and realised false admissions per campaign at label thresholds of 0.010, 0.015 (registered), and 0.020.

Referee	Controller	Yield at 0.010 / 0.015 / 0.020	False Admissions at 0.010 / 0.015 / 0.020
Frozen	Round-Robin	0.403 / 0.403 / 0.400	5.4 / 11.7 / 13.8
Frozen	LLM (Gemini)	0.471 / 0.471 / 0.466	5.7 / 11.7 / 14.7
Peeking	Round-Robin	0.407 / 0.406 / 0.402	78.5 / 86.2 / 90.2
Peeking	LLM (Gemini)	0.546 / 0.546 / 0.532	31.9 / 37.4 / 45.7
Adaptive Threshold	Round-Robin	0.390 / 0.390 / 0.387	97.1 / 105.2 / 108.2
Adaptive Threshold	LLM (Gemini)	0.478 / 0.478 / 0.476	37.5 / 43.5 / 47.0
No Gate	Round-Robin	0.370 / 0.367 / 0.361	185.8 / 196.0 / 200.5
No Gate	LLM (Gemini)	0.596 / 0.585 / 0.547	66.7 / 78.0 / 95.8

scale. The profiles rank families correctly and understate only the level. The referee never reads the profile. The controller uses it only to weigh families against one another.

Appendix E Data, point-in-time rules and the factor library

Every quantity used on a day is known before that day’s position is taken. Membership uses the latest index-weight snapshot dated at or before the previous trading day. Prices are adjusted by the vendor’s cumulative factor. Halts are missing values. A name that closed at its price limit ($|\Delta p| \geq 9.5\%$) on day $t - 1$ is excluded from the day- t universe. Fundamentals become visible on their announcement date and have a 400-day staleness cut. The panel covers 2834 trading days. It includes 1356 stocks that were ever members. Every vendor-extract partition has a SHA-256 in a provenance manifest.

The factor library. The library is fixed in advance, keeping the hypothesis universe finite as Proposition 1 requires. Each family represents an economic hypothesis with a fixed sign and parameter grid. Each

Table 12: Four-family library results show realised false admissions under three labels. Values are means per campaign over four starts $\times$ five seeds. *Full* is the registered label. *After* judges each admitted factor only on days after admission. δ_f is each family’s break-even (liquidity 0.012, volatility 0.017, momentum 0.031, reversal 0.074).

Referee	Controller	Admitted	False: Full	False: After	False at δ_f	Edge After	Yield: Full / After
Frozen	Round-Robin	214.7	11.7	17.6	96.0	0.032	0.403 / 0.380
Frozen	Bandit	260.8	9.4	14.4	114.3	0.032	0.474 / 0.460
Frozen	LLM (Gemini)	254.8	11.7	15.8	111.7	0.032	0.466 / 0.458
Peeking	Round-Robin	312.4	86.2	91.2	180.4	0.023	0.406 / 0.390
Peeking	Bandit	342.8	38.4	52.6	170.5	0.027	0.495 / 0.457
Peeking	LLM (Gemini)	354.6	37.4	41.5	152.9	0.029	0.545 / 0.511
Adaptive Threshold	Round-Robin	322.3	105.2	107.5	197.4	0.021	0.390 / 0.384
Adaptive Threshold	Bandit	327.6	51.0	54.4	185.8	0.027	0.442 / 0.430
Adaptive Threshold	LLM (Gemini)	330.2	43.5	44.4	158.2	0.029	0.478 / 0.466
No Gate	Round-Robin	398.0	196.0	195.6	276.2	0.016	0.367 / 0.358
No Gate	Bandit	398.0	189.8	194.1	265.6	0.015	0.372 / 0.352
No Gate	LLM (Gemini)	394.0	77.2	77.0	185.3	0.027	0.577 / 0.565

Table 13: Nine-family library results show the same measures. They exclude a per-family break-even because it is undefined for the five fundamental families.

Referee	Controller	Admitted	False: Full	False: After	Edge After	Yield: Full / After
Frozen	Round-Robin	127.2	9.8	16.6	0.028	0.246 / 0.225
Frozen	Bandit	188.1	11.9	17.6	0.030	0.357 / 0.340
Frozen	LLM (Gemini)	198.6	14.6	20.3	0.030	0.377 / 0.364
Peeking	Round-Robin	256.9	127.3	131.8	0.016	0.244 / 0.233
Peeking	Bandit	295.8	97.0	107.2	0.020	0.360 / 0.330
Adaptive Threshold	Round-Robin	296.4	175.2	174.9	0.013	0.230 / 0.228
Adaptive Threshold	Bandit	299.6	128.8	129.7	0.019	0.292 / 0.285
No Gate	Round-Robin	398.0	295.6	292.9	0.009	0.192 / 0.189
No Gate	Bandit	398.0	293.8	291.1	0.008	0.194 / 0.190

submission draws one specification from that grid (Table 26).

Table 14: Four-family library results show fixed-horizon comparator referees run as counterfactual programs on the frozen cell’s recorded submission stream. Values are means per campaign over four starts $\times$ five seeds. Each tests the post-submission mean once, 500 or 250 days after submission. Each applies Benjamini–Hochberg at 0.05 whenever a window closes (*Daily*, deployable) or once at the campaign’s end (*At End*, so no yield or wait).

Controller	Referee	Admitted	False: Full	False: After	Share False	Edge After	Yield	Wait
Round-Robin	Frozen (Recorded)	214.7	11.7	17.6	0.08	0.032	0.403	439
Round-Robin	Fixed 500 (Daily)	208.0	16.4	22.2	0.11	0.029	0.358	500
Round-Robin	Fixed 500 (At End)	207.2	16.0	21.8	0.11	0.029	–	–
Round-Robin	Fixed 250 (Daily)	187.4	16.6	19.4	0.10	0.030	0.314	250
Bandit	Frozen (Recorded)	260.8	9.4	14.4	0.06	0.032	0.474	408
Bandit	Fixed 500 (Daily)	262.6	12.2	20.4	0.08	0.028	0.441	500
Bandit	Fixed 500 (At End)	262.3	12.1	20.2	0.08	0.028	–	–
Bandit	Fixed 250 (Daily)	239.6	9.8	13.3	0.06	0.031	0.404	250

Table 15: Nine-family library results show the same comparators.

Controller	Referee	Admitted	False: Full	False: After	Share False	Edge After	Yield	Wait
Round-Robin	Frozen (Recorded)	127.2	9.8	16.6	0.12	0.028	0.246	543
Round-Robin	Fixed 500 (Daily)	150.2	41.4	49.7	0.33	0.021	0.214	500
Round-Robin	Fixed 500 (At End)	148.8	40.8	48.9	0.33	0.021	–	–
Round-Robin	Fixed 250 (Daily)	115.0	30.2	33.8	0.29	0.023	0.159	250
Bandit	Frozen (Recorded)	188.1	11.9	17.6	0.09	0.030	0.357	436
Bandit	Fixed 500 (Daily)	207.8	33.2	44.0	0.22	0.024	0.330	500
Bandit	Fixed 500 (At End)	207.3	33.2	43.8	0.21	0.024	–	–
Bandit	Fixed 250 (Daily)	177.2	25.3	29.2	0.17	0.027	0.287	250

Table 16: Net Sharpe of counterfactual books on the round-robin frozen cell’s candidate sequence. Results use the four-family library and five seeds per start. They report means with standard errors and differences paired per seed. *Retirement off* holds every factor admitted by the frozen referee through the campaign’s end. *Fixed 500 (Daily)* is Table 14’s deployable 500-day comparator. It is held from its admission day until the runs’ e-detector retires it.

Start	No Gate	Frozen	Frozen, Retirement Off	Retirement Off Minus Frozen	Fixed 500 (Daily)	Fixed 500 Minus Frozen
2016	+0.74 $\pm$ 0.02	+0.40 $\pm$ 0.03	+0.41 $\pm$ 0.03	+0.00 $\pm$ 0.00	+0.60 $\pm$ 0.01	+0.20 $\pm$ 0.03
2017	+0.78 $\pm$ 0.01	+0.46 $\pm$ 0.01	+0.47 $\pm$ 0.01	+0.01 $\pm$ 0.00	+0.59 $\pm$ 0.02	+0.13 $\pm$ 0.02
2018	+0.75 $\pm$ 0.01	+0.42 $\pm$ 0.02	+0.42 $\pm$ 0.02	+0.01 $\pm$ 0.00	+0.65 $\pm$ 0.02	+0.24 $\pm$ 0.02
2019	+0.62 $\pm$ 0.04	+0.49 $\pm$ 0.05	+0.50 $\pm$ 0.05	+0.01 $\pm$ 0.00	+0.61 $\pm$ 0.02	+0.12 $\pm$ 0.04

Table 17: Controller contrasts on the four-family library under the frozen referee, split by candidates submitted before and from 2020-01-02. Each campaign has about 114 and 308 such candidates. LLM-minus-baseline differences are paired by start and seed. Values are means $\pm$ s.e. over 20 pairs.

Family	Contrast	Yield Diff. Pre-2020	Yield Diff. Post-2020	False Adm. Diff. Pre-2020	False Adm. Diff. Post-2020
Gemini	LLM – Round-Robin	+0.009 $\pm$ 0.005	+0.088 $\pm$ 0.004	−0.2 $\pm$ 0.1	+0.1 $\pm$ 0.2
Gemini	LLM – Bandit	+0.026 $\pm$ 0.006	−0.016 $\pm$ 0.004	+0.3 $\pm$ 0.2	+1.9 $\pm$ 0.7
DeepSeek	LLM – Round-Robin	+0.028 $\pm$ 0.009	+0.097 $\pm$ 0.004	−0.1 $\pm$ 0.1	+1.1 $\pm$ 0.6
DeepSeek	LLM – Bandit	+0.045 $\pm$ 0.011	−0.007 $\pm$ 0.004	+0.5 $\pm$ 0.2	+2.8 $\pm$ 0.9
GPT	LLM – Round-Robin	+0.008 $\pm$ 0.003	+0.050 $\pm$ 0.004	−0.1 $\pm$ 0.1	+1.7 $\pm$ 0.4
GPT	LLM – Bandit	+0.025 $\pm$ 0.006	−0.054 $\pm$ 0.003	+0.5 $\pm$ 0.2	+3.4 $\pm$ 0.8

Table 18: Retirement outcomes on real data using the four-family library, frozen referee, and round-robin controller. Results cover four starts $\times$ five seeds. They report factors retired per campaign and family. For retired factors, results also report mean daily rank-IC during the held window and the remaining campaign. These factors had at least 21 days held and 21 days left in the campaign.

Family	Retired	Share Retired	Days Held	IC While Held	IC After	After $< \delta$	After < 0
Liquidity	12.1	0.24	207	−0.0053	+0.0040	1.00	0.05
Momentum	0.5	1.00	354	−0.0131	+0.0035	0.89	0.22
Reversal	31.8	0.36	838	+0.0145	+0.0338	0.00	0.00
All	44.3	0.21	661	+0.0088	+0.0254	0.28	0.01

Table 19: Net Sharpe of counterfactual books under the canonical construction. The books are frozen, ungated, (a) the referee’s factors on the ungated schedule, and (b) a fixed random subset of ungated factors matching the referee’s count. *Breadth* is no gate less (b). *Selection* is (b) less (a). *Earliness* is (a) less frozen. The three sum to the gap. Sharpe here uses 252 days.

Start	Frozen	No Gate	(a)	(b)	Breadth	Selection	Earliness
2016	+0.40	+0.74	+0.63	+0.72	+0.02	+0.09	+0.23
2017	+0.46	+0.78	+0.65	+0.75	+0.04	+0.10	+0.19
2018	+0.42	+0.75	+0.67	+0.71	+0.04	+0.04	+0.25
2019	+0.49	+0.62	+0.53	+0.57	+0.05	+0.04	+0.04

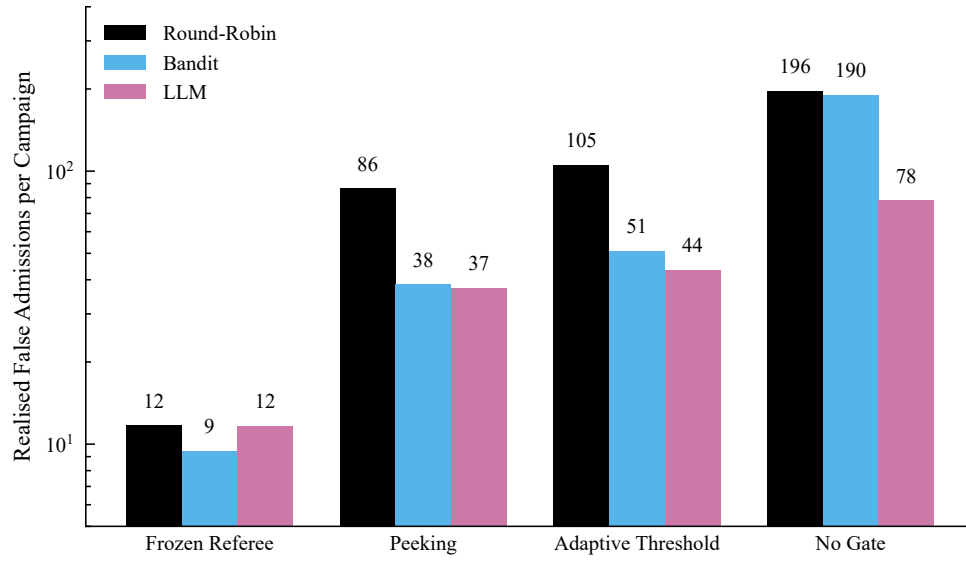


Figure 6: Realised false admissions per campaign by referee arm and controller (real data, four-family library; means over four start years $\times$ five seeds; log scale). The referee sets the count; the controller moves it by a factor of 2–2.5, and only when the referee leaks.

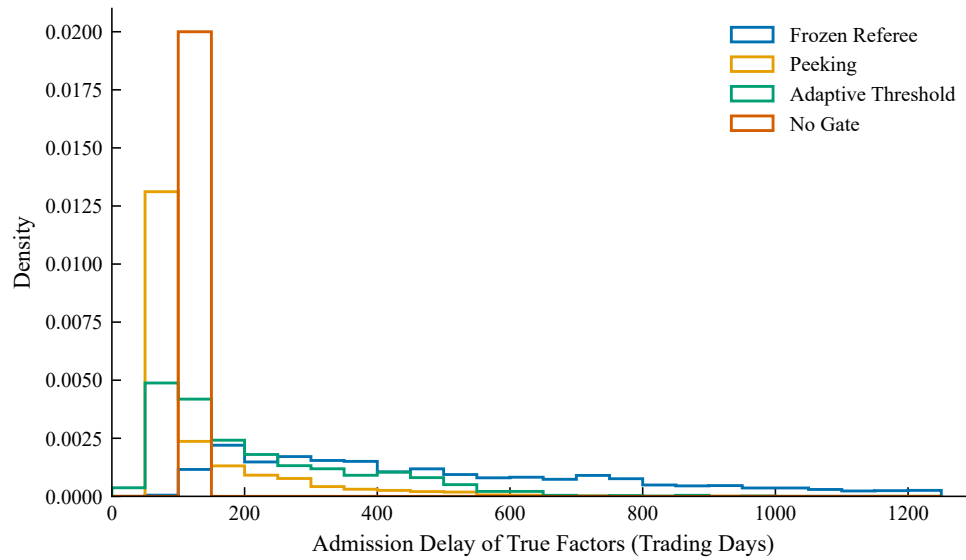


Figure 7: Admission delay of realised-true factors by referee arm (LLM controller, four-family library, 2016 start, five seeds pooled). The no-gate spike is the pre-registered 126-day incubation.

Table 20: Nine-family library with the round-robin controller, 2016 start, and five pooled seeds. The table reports sleeves ever held by the book. It also reports each family’s mean first-held campaign day, of 2,589. Finally, it reports the share of holding windows with midpoints after 2020-01-02, campaign day 975.

Family	Certified: Sleeves	First Held (Mean Day)	After 2020	Ungated: Sleeves	First Held (Mean Day)	After 2020
Growth	42	1151	100%	235	1365	83%
Investment	0			412	1419	76%
Liquidity	126	1511	100%	213	1264	83%
Momentum	0			354	1561	84%
Quality	45	1177	100%	306	1541	86%
Reversal	237	1722	100%	164	1266	100%
Size	2	1842	100%	321	1242	65%
Value	171	1881	99%	184	1151	68%
Volatility	203	1509	100%	151	1190	100%
All Families	826	1612	100%	2340	1366	81%

Table 21: Paired realised-yield contrasts under the frozen referee on the four-family library. Contrasts compare the LLM controller with round-robin and the bandit. They report the mean difference across five common-random-number seeds and its standard error. They also report the Holm-adjusted two-sided p within the model family, over 24 tests.

Family	Start	LLM – Round-Robin			LLM – Bandit Yield	Holm p
		Yield	Holm p			
DeepSeek	2016	+0.095 $\pm$ 0.007	0.006	+0.039 $\pm$ 0.007	0.124	
DeepSeek	2017	+0.079 $\pm$ 0.005	0.005	+0.007 $\pm$ 0.004	1.000	
DeepSeek	2018	+0.080 $\pm$ 0.010	0.034	–0.004 $\pm$ 0.006	1.000	
DeepSeek	2019	+0.070 $\pm$ 0.006	0.012	–0.002 $\pm$ 0.007	1.000	
Gemini	2016	+0.064 $\pm$ 0.009	0.122	+0.008 $\pm$ 0.007	1.000	
Gemini	2017	+0.070 $\pm$ 0.005	0.008	–0.002 $\pm$ 0.002	1.000	
Gemini	2018	+0.076 $\pm$ 0.003	0.001	–0.009 $\pm$ 0.003	1.000	
Gemini	2019	+0.063 $\pm$ 0.006	0.033	–0.009 $\pm$ 0.012	1.000	
GPT	2016	+0.046 $\pm$ 0.005	0.039	–0.010 $\pm$ 0.005	1.000	
GPT	2017	+0.038 $\pm$ 0.004	0.029	–0.034 $\pm$ 0.006	0.156	
GPT	2018	+0.045 $\pm$ 0.008	0.136	–0.039 $\pm$ 0.005	0.060	
GPT	2019	+0.028 $\pm$ 0.008	0.537	–0.043 $\pm$ 0.007	0.111	

Table 22: The same contrasts on the nine-family library.

Family	Start	LLM – Round-Robin			LLM – Bandit Yield	Holm p
		Yield	Holm p			
DeepSeek	2016	+0.142 $\pm$ 0.004	0.000	+0.029 $\pm$ 0.007	0.178	
DeepSeek	2017	+0.120 $\pm$ 0.009	0.003	+0.011 $\pm$ 0.008	1.000	
DeepSeek	2018	+0.148 $\pm$ 0.007	0.001	+0.026 $\pm$ 0.004	0.072	
DeepSeek	2019	+0.090 $\pm$ 0.009	0.009	–0.010 $\pm$ 0.005	1.000	
Gemini	2016	+0.163 $\pm$ 0.003	0.000	+0.050 $\pm$ 0.005	0.012	
Gemini	2017	+0.134 $\pm$ 0.007	0.001	+0.026 $\pm$ 0.006	0.260	
Gemini	2018	+0.159 $\pm$ 0.013	0.005	+0.036 $\pm$ 0.010	0.310	
Gemini	2019	+0.092 $\pm$ 0.015	0.068	–0.009 $\pm$ 0.012	1.000	
GPT	2016	+0.059 $\pm$ 0.010	0.066	–0.054 $\pm$ 0.007	0.032	
GPT	2017	+0.052 $\pm$ 0.003	0.001	–0.057 $\pm$ 0.009	0.064	
GPT	2018	+0.051 $\pm$ 0.010	0.116	–0.072 $\pm$ 0.008	0.021	
GPT	2019	+0.018 $\pm$ 0.010	1.000	–0.082 $\pm$ 0.006	0.004	

Table 23: Coverage of the real-data factorial. Every row covers four start years and five seeds. The last row is the confirmatory extension run after the factorial.

Controllers	Model Families	Referee Arms	Libraries	Starts $\times$ Seeds	Cells
Round-Robin, Bandit LLM	None	All Four	Both	4×5	320
LLM	Gemini, DeepSeek, GPT Gemini	Frozen	Both	4×5	120
LLM		Peeking, Adaptive Threshold, No Gate	Four-Family	4×5	60
LLM Total	DeepSeek, GPT	No Gate	Four-Family	4×5	40 540

Table 24: Dated amendments to the pre-registered design. A5 and A9 were never issued. Every book amendment used zero-cost re-booking of the recorded cells (Proposition 2).

Id	Date	What Changed	Affects
A1	2026-09-02	LLM costs metered at billed prices; spend cap \$335	No (accounting)
A2	2026-09-02	Deterministic iteration order; longer retry window	No (engineering)
A3	2026-09-03	Escalation-price sensitivity computed by re-scoring recorded episodes	No (analysis)
A4	2026-09-04	Abstract effort pins realised through the measured vendor map	No (runtime)
A6	2026-09-04	Probe environment audited; two world defects fixed (world v2)	Probe Environment
A7	2026-09-05	Real-data world specification on the CSI 500	Real-Data World
A8	2026-09-07	Nine-family library; trading-frequency slice	Library
A10	2026-09-07	Framework consolidation before the restart; earlier recorded runs not used	All Real-Data Runs
A10b	2026-09-07	Ex-ante book controls (turnover cap, index hedge); canonical constructed book	Book
A11	2026-09-11	Certification, persistence and execution separated; per-sleeve calibrated holding period	Book
A11b	2026-09-11	Five-frequency calendar grid; dynamic re-calibration with a cool-down	Book
A11c	2026-09-13	Conversion rate fixed as a market constant after a ledger audit of the first pass	Book

Table 25: The controller ladder. Every rung uses the same library, submission budget and referee. The rungs differ in policy and, at the top rung, inputs and actions. Thus, the contrasts compare controller packages.

Rung	Allocation Across Families	Response to an Alarm	Instruments	Inputs Beyond Family Statistics
Round-robin (C0)	Cycles through families	Scripted: retire and refill the same family	Fixed menu	None
Discounted-UCB bandit (C1)	Reward per family: admissions less retirements, discounted	Scripted, as C0	Fixed menu	None
LLM controller (C2)	Reads statistics, family profiles and memory	Diagnosis (idiosyncratic death, crowding, regime shift, data breakage) or escalation; chooses the response	Authors, tests and mounts probes	Family profiles, memory, the reasons for retirements

Table 26: The nine factor families (the four-family library is the first four). Prices use adjusted closes. Fundamentals are announcement-aligned. The grid is enumerable, fixing the hypothesis universe before the campaign.

Family	Signal (sign)	Parameters Drawn	Specs
Momentum	$\log(P_{t-1-s}/P_{t-1-s-L})$ (+)	$L \in [40, 250]$, skip $s \in [0, 21]$	4,642
Short-Horizon Reversal	$-\log(P_{t-1}/P_{t-1-L})$ (+ after sign)	$L \in [3, 30]$; turnover-scaled or not	56
Volatility	$-(\text{std, downside std or max})$ of daily returns over L (+ after sign)	$L \in [10, 130]$; three kinds	363
Liquidity	$-\text{mean turnover over } L$, or Amihud illiquidity (+ after sign)	$L \in [10, 130]$; two kinds	242
Value	Book, earnings, sales or dividends to price, smoothed over L (+)	$L \in [1, 21]$; four kinds	84
Size	$-\log$ market capitalisation, circulating or total (+ after sign)	$L \in [1, 21]$; two kinds	42
Quality	ROE, quarterly ROE, ROA, gross or net margin (+)	$L \in [1, 63]$; five kinds	315
Growth	Net profit, revenue or EPS growth, year on year (+)	$L \in [1, 63]$; three kinds	189
Investment (asset Growth)	Asset or equity growth, year on year (−)	$L \in [1, 63]$; two kinds	126